



EEC 431

ELECTROMAGNETIC FIELD THEORY

HIGHER NATIONAL DIPLOMA (HND) II

LECTURE NOTES

Organised by:

Dr. Aliyu A. B. Mayobani

PgCLTHE MSc. Ph.D. AFHEA FHEA MIET MIEEE

Department of Electrical & Electronics

School of Engineering, Federal Polytechnic Mubi PMB 35,

Adamawa State, Nigeria.



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COURSE BRIEF

Electromagnetic field theory: is concerned with the study of charges at rest and in motion.

Electromagnetic principles are fundamental to the study of electrical engineering.

Electromagnetic theory is also required for the understanding, analysis and design of various electrical, electromechanical and electronic systems.

Electromagnetic theory can be thought of as generalization of circuit theory.

Electromagnetic theory deals directly with the electric and magnetic field vectors whereas circuit theory deals with the voltages and currents. Voltages and currents are integrated effects of electric and magnetic fields respectively.

Electromagnetic field problems involve three space variables along with the time variable and hence the solution tends to become correspondingly complex. Vector analysis is the required mathematical tool with which electromagnetic concepts can be conveniently expressed and best comprehended. Since use of vector analysis in the study of electromagnetic field theory is prerequisite, first we will go through vector algebra.

Applications of Electromagnetic theory:

This module basically consists of static electric fields, static magnetic fields, time-varying fields & its applications.

One of the most common applications of electrostatic fields is the deflection of a charged particle such as an electron or proton in order to control its trajectory. The deflection is achieved by maintaining a potential difference between a pair of parallel plates. This principle is used in CROs, ink-jet printer etc. Electrostatic fields are also used for sorting of minerals for example in ore separation. Other applications are in electrostatic generator and electrostatic voltmeter. The most common applications of static magnetic fields are in dc machines. Other applications include magnetic deflection, magnetic separator, cyclotron, hall effect sensors, magneto hydrodynamic generator etc.

**Recommended Textbooks:**

- [1]. W. H. Hayt (Jr), J. A. Buck, “Engineering Electromagnetics”, TMH
- [2]. K. E. Lonngren, S.V. Savor, “Fundamentals of Electromagnetics with Matlab”, PHI
- [3]. E.C.Jordan, K.G. Balmain, “Electromagnetic Waves & Radiating System”, PHI.
- [4]. M. N. Sadiku, “Elements of Electromagnetics”, Oxford University Press.
- [5] D. Tougaw “Applied Electromagnetic Field Theory” Valparaiso University Press



INTRODUCTION

Vector Analysis:

The quantities that we deal in electromagnetic theory may be either **scalar** or **vectors**.

Scalars are quantities characterized by magnitude only. A quantity that has direction as well as magnitude is called a vector. In electromagnetic theory both scalar and vector quantities are function of *time* and *position*.

A vector $\vec{A}$ can be written as, $\vec{A} = \hat{a} A$, where, $A = |\vec{A}|$ is the magnitude and $\hat{a} = \frac{\vec{A}}{|\vec{A}|}$ is the unit vector which has unit magnitude and same direction as that of $\vec{A}$.

Two vectors $\vec{A}$ and $\vec{B}$ are added together to give another vector $\vec{C}$. Thus,

$$\vec{C} = \vec{A} + \vec{B} \quad (1.1)$$

Let us see the manipulations (for addition) of two vectors, which has two rules: the

Parallelogram law and the **Head & tail rule**.

Scaling of a vector is defined as $\vec{C} = \alpha \vec{B}$, where $\vec{C}$ is scaled version of vector $\vec{B}$ and α is a scalar. Some important laws of vector algebra are:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \quad \dots \quad (\text{Commutative Law}) \quad (1.3)$$

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} \quad \dots \quad (\text{Associative Law}) \quad (1.4)$$

$$\alpha(\vec{A} + \vec{B}) = \alpha\vec{A} + \alpha\vec{B} \quad \dots \quad (\text{Distributive Law}) \quad (1.5)$$

The position vector $\vec{r}_P$ of a point P is the directed distance from the origin (O) to P ,

i.e., $\vec{r}_P = \vec{OP}$. If $\vec{r}_P = \vec{OP}$ and $\vec{r}_Q = \vec{OQ}$ are the position vectors of the points P and Q then the distance vector

$$\vec{PQ} = \vec{OQ} - \vec{OP} = \vec{r}_Q - \vec{r}_P$$

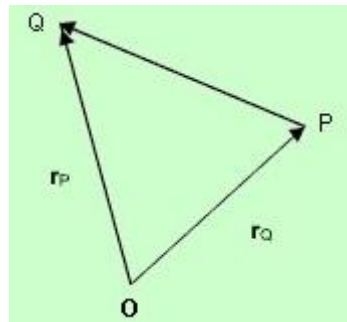


Fig 1.3: Distance Vector

Product of Vectors

When two vectors $\vec{A}$ and $\vec{B}$ are multiplied, the result is either a scalar or a vector depending how the two vectors were multiplied. The two types of vector multiplication are:

Scalar product (or dot product) $\vec{A} \cdot \vec{B}$ gives a scalar.

Vector product (or cross product) $\vec{A} \times \vec{B}$ gives a vector.

The dot product between two vectors is defined as $\vec{A} \cdot \vec{B} = |A||B|\cos\theta_{AB}$ (1.6)

Vector product

$$\vec{A} \times \vec{B} = |A||B|\sin\theta_{AB} \cdot \vec{n}$$

$\vec{n}$ is unit vector perpendicular to $\vec{A}$ and $\vec{B}$

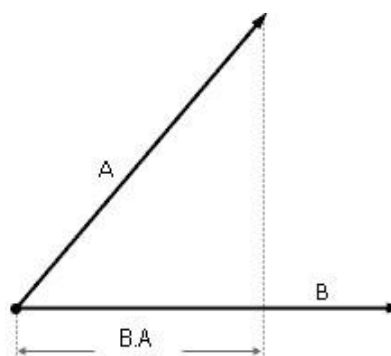


Fig 1.4 : Vector dot product

The dot product is commutative i.e., $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ and distributive i.e., $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$

Associative law does not apply to scalar product.

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The vector or cross product of two vectors $\vec{A}$ and $\vec{B}$ is denoted by $\vec{A} \times \vec{B}$. $\vec{A} \times \vec{B}$ is denoted by perpendicular to the plane containing $\vec{A}$ and $\vec{B}$, the magnitude is given by $|\vec{A}||\vec{B}|\sin \theta_{AB}$ and direction is given by right hand rule.

$$\vec{A} \times \vec{B} = \hat{a}_n AB \sin \theta_{AB} \quad (1.7)$$

where $\hat{a}_n$ is the unit vector given by,

$$\hat{a}_n = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|}$$

The following relations hold for vector product.

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A} \quad \text{i.e., cross product is non-commutative} \quad (1.8)$$

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C} \quad \text{i.e., cross product is distributive} \quad (1.9)$$

$$\vec{A} \times (\vec{B} \times \vec{C}) \neq (\vec{A} \times \vec{B}) \times \vec{C} \quad \text{i.e., cross product is non associative} \quad (1.10)$$

Scalar and vector triple product:

$$\text{Scalar triple product} \quad \vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B}) \quad (1.11)$$

$$\text{Vector triple product} \quad \vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B}) \quad (1.12)$$

Co-ordinate Systems:

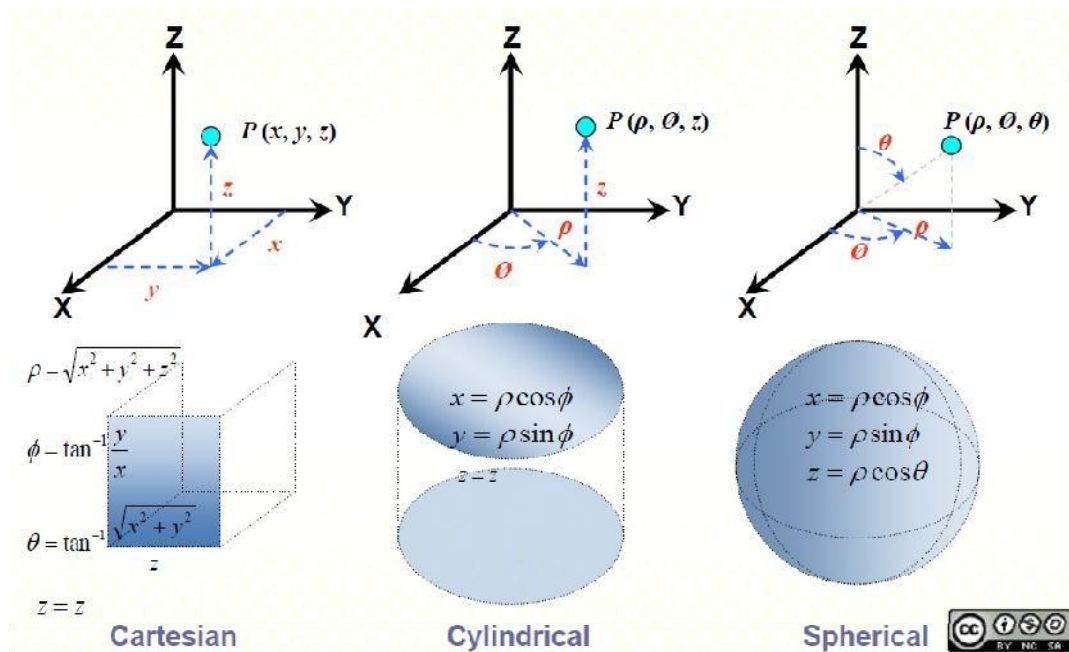
In order to describe the spatial variations of the quantities, we require using appropriate co-ordinate system. A point or vector can be represented in an orthogonal coordinate system. An orthogonal system is one in which the co-ordinates are mutually perpendicular.

In electromagnetic theory many physical quantities are vectors, which are having different components. So we use orthogonal co-ordinate systems for representing those quantities and depending on the symmetry of the physical quantities different coordinate systems are used.



The electromagnetic (static) problem cases

Coordinate	The problems most often encountered	
	Electrostatic Cases	Magnetostatic Cases
Cartesian	- Any uniform charge distribution using a scale of cartesian coordinates.	- The current flowing on the horizontal infinite extent plane.
Cylindrical	- Uniform charge distribution on cylindrical conductor. - Uniform charge distribution on the infinite length of line. - Uniform charge distribution on the infinite extent horizontal plane. - Uniform charge distribution on the horizontal circle plane. - Uniform charge distribution on the circle line.	- The current flowing in circumference. - The current flowing in an infinite straight line. - The current flows in the solenoid and toroid.
Spherical	- Uniform charge distribution on the surface of a sphere. - Uniform charge distribution of the point.	- Problem for the case of spheres less found.



**Del Operator:**

Del is a vector differential operator. The del operator will be used in for differential operations throughout any course on field theory. The following equation is the del operator for different coordinate systems.

$$\nabla = \frac{\partial}{\partial x} \hat{a}_x + \frac{\partial}{\partial y} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z = \nabla_{x,y,z}$$

$$\nabla = \frac{\partial}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial}{\partial \phi} \hat{a}_\phi + \frac{\partial}{\partial z} \hat{a}_z$$

$$\nabla = \frac{\partial}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \hat{a}_\phi$$

Gradient of a Scalar:

- The gradient of a scalar field, V , is a vector that represents both the magnitude and the direction of the maximum space rate of increase of V .

$$\nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z = \nabla V_{x,y,z}$$

- To help visualize this concept, take for example a topographical map. Lines on the map represent equal magnitudes of the scalar field. The gradient vector crosses map at the location where the lines packed into the most dense space and perpendicular (or normal) to them. The orientation (up or down) of the gradient vector is such that the field is increased in magnitude along that direction.

Fundamental properties of the gradient of a scalar field

- The magnitude of gradient equals the maximum rate of change in V per unit distance
- Gradient points in the direction of the maximum rate of change in V
- Gradient at any point is perpendicular to the constant V surface that passes through that point – The projection of the gradient in the direction of the unit vector $\mathbf{a}$, is

$$\nabla V \cdot \hat{\mathbf{a}}$$

and is called the directional derivative of V along $\mathbf{a}$. This is the rate of change of V in the direction of $\mathbf{a}$.

- If $\mathbf{A}$ is the gradient of V , then V is said to be the scalar potential of $\mathbf{A}$.

**Del operator:**

Del is a vector differential operator. The del operator will be used in for differential operations throughout any course on field theory. The following equation is the del operator for different coordinate systems.

$$\nabla = \frac{\partial}{\partial x} \hat{a}_x + \frac{\partial}{\partial y} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z = \nabla_{x,y,z}$$

$$\nabla = \frac{\partial}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial}{\partial \phi} \hat{a}_\phi + \frac{\partial}{\partial z} \hat{a}_z$$

$$\nabla = \frac{\partial}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \hat{a}_\phi$$

Gradient of a Scalar:

- The gradient of a scalar field, V, is a vector that represents both the magnitude and the direction of the maximum space rate of increase of V.

$$\nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z = \nabla V_{x,y,z}$$

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-Fundamental properties of the gradient of a scalar field

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$$\nabla V \cdot \hat{a}$$

and is called the directional derivative of V along **a**. This is the rate of change of V in the direction of **a**.

- If **A** is the gradient of V, then V is said to be the scalar potential of **A**.

**Divergence of a Vector:**

- The divergence of a vector, $\mathbf{A}$, at any given point P is the outward flux per unit volume as volume shrinks about P.

$$\text{div} \vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

Divergence Theorem:

- The divergence theorem states that the total outward flux of a vector field, $\mathbf{A}$, through the closed surface, S, is the same as the volume integral of the divergence of $\mathbf{A}$.
- This theorem is easily shown from the equation for the divergence of a vector field.

$$\begin{aligned}\vec{A} &= A_1 \hat{a}_1 + A_2 \hat{a}_2 + A_3 \hat{a}_3 \\ \text{div} \vec{A} &= \nabla \cdot \vec{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v} \\ \int_v \nabla \cdot \vec{A} dv &= \oint_s \vec{A} \cdot d\vec{S}\end{aligned}$$

Curl of a Vector:

- The curl of a vector, $\mathbf{A}$ is an axial vector whose magnitude is the maximum circulation of $\mathbf{A}$ per unit area as the area tends to zero and whose direction is the normal direction of the area when the area is oriented to make the circulation maximum.
- Curl of a vector in each of the three primary coordinate systems are,



Cartesian $\nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \hat{a}_x - \left[\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right] \hat{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \hat{a}_z$

Cylindrical $\nabla \times \vec{A} = \frac{1}{\rho} \begin{vmatrix} \hat{a}_\rho & \rho \hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix} = \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \hat{a}_\rho - \left[\frac{\partial A_z}{\partial \rho} - \frac{\partial A_\rho}{\partial z} \right] \hat{a}_\phi + \frac{1}{\rho} \left[\frac{\partial(\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right] \hat{a}_z$

Spherical $\nabla \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{a}_r & r \hat{a}_\theta & r \sin \theta \hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$

$$\nabla \times \vec{A} = \frac{1}{r \sin \theta} \left[\frac{\partial(A_\phi \sin \theta)}{\partial \theta} - \frac{\partial A_\theta}{\partial \phi} \right] \hat{a}_r - \frac{1}{r} \left[\frac{\partial(r A_\phi)}{\partial r} - \frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} \right] \hat{a}_\theta + \frac{1}{r} \left[\frac{\partial(r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right] \hat{a}_\phi$$

Stokes Theorem:

Stokes theorem states that the circulation of a vector field $\mathbf{A}$, around a closed path, L is equal to the surface integral of the curl of $\mathbf{A}$ over the open surface S bounded by L . This theorem has been proven to hold as long as $\mathbf{A}$ and the curl of $\mathbf{A}$ are continuous along the closed surface S of a closed path L

- This theorem is easily shown from the equation for the curl of a vector field.

$$\vec{A} = A_1 \hat{a}_1 + A_2 \hat{a}_2 + A_3 \hat{a}_3$$

$$\text{curl} \vec{A} = \nabla \times \vec{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \vec{A} \cdot d\vec{l}}{\Delta S} \right)_{\max} \hat{a}_n$$

$$\oint_L \vec{A} \cdot d\vec{l} = \oint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

Examples: 1

$$\mathbf{A} = \hat{a}_x + 2\hat{a}_y + 3\hat{a}_z$$

1. Given that

$$\mathbf{B} = -2\hat{a}_x + \hat{a}_y$$

- Determine the angle between the vectors $\mathbf{A}$ and $\mathbf{B}$.



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- b. Find the unit vector which is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.

Solution:

- a. We know that for two given vector $\mathbf{A}$ and $\mathbf{B}$,

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta$$

For the two vectors $\mathbf{A}$ and $\mathbf{B}$

$$|\mathbf{A}| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

$$|\mathbf{B}| = \sqrt{(-2)^2 + 1} = \sqrt{5}$$

$$\therefore \sqrt{14} \cdot \sqrt{5} \cos \theta = 1 \quad \text{or} \quad \theta = \cos^{-1} \left(\frac{1}{\sqrt{70}} \right)$$

- b. We know that is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.

$$\begin{aligned} \therefore \mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 1 & 2 & 3 \\ -2 & 0 & 1 \end{vmatrix} \\ &= 2\hat{a}_x - (1+6)\hat{a}_y + 4\hat{a}_z \end{aligned}$$

The unit vector $\hat{n}$ perpendicular to both $\mathbf{A}$ and $\mathbf{B}$ is given by,

$$\hat{n} = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} = \frac{2\hat{a}_x - 7\hat{a}_y + 4\hat{a}_z}{\sqrt{2^2 + (-7)^2 + 4^2}} = \frac{2\hat{a}_x - 7\hat{a}_y + 4\hat{a}_z}{\sqrt{69}}$$

Examples: 2

Given the vectors

$$\mathbf{A} = \hat{a}_x + 2\hat{a}_y + 5\hat{a}_z$$

$$\mathbf{B} = 5\hat{a}_x - \hat{a}_y + 3\hat{a}_z$$

Find:

- The vector $\mathbf{C} = \mathbf{A} + \mathbf{B}$ at a point $P(0, 2, -3)$.
- The component of $\mathbf{A}$ along $\mathbf{B}$ at P .

**Solution:**

The vector B is cylindrical coordinates.

This vector in Cartesian coordinate can be written as:

$$\mathbf{B} = B_x \hat{a}_x + B_y \hat{a}_y + B_z \hat{a}_z$$

Where

$$B_x = \mathbf{B} \cdot \hat{a}_x = 5 \hat{a}_\rho \cdot \hat{a}_x - \hat{a}_\phi \cdot \hat{a}_x + 3 \hat{a}_z \cdot \hat{a}_x$$

$$= 5 \cos \phi + \sin \phi$$

$$B_y = \mathbf{B} \cdot \hat{a}_y = 5 \sin \phi - \cos \phi$$

$$B_z = \mathbf{B} \cdot \hat{a}_z$$

The point P(0,2,-3) is in the y-z plane for which $\phi = \frac{\pi}{2}$.

$$\therefore \mathbf{B} = \hat{a}_x + 5\hat{a}_y + 3\hat{a}_z$$

a. $\mathbf{C} = \mathbf{A} + \mathbf{B}$

$$\begin{aligned} &= \left(\hat{a}_x + 2\hat{a}_y + 5\hat{a}_z \right) + \left(\hat{a}_x + 5\hat{a}_y + 3\hat{a}_z \right) \\ &= 2\hat{a}_x + 7\hat{a}_y + 8\hat{a}_z \end{aligned}$$

b. Component of $\mathbf{A}$ along $\mathbf{B}$ is $|\mathbf{A}| \cos \theta$ where θ is the angle between A and B.

i.e., $\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{B}|} = \frac{1+10+15}{\sqrt{1+25+9}} = \frac{26}{\sqrt{35}} =$

Examples: 3

A vector field is given by

$$\mathbf{A} = \rho \cos \phi \hat{a}_\rho + \rho z \sin \phi \hat{a}_z$$



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Transform this vector into rectangular co-ordinates and calculate its magnitude at P(1,0,1).

Solution:

$$\text{Given, } \mathbf{A} = \rho \cos \phi \hat{a}_\rho + \rho z \sin \phi \hat{a}_z$$

The components of the vector in Cartesian coordinates can be computed as follows:

$$\begin{aligned} A_x &= A \hat{a}_x \\ &= \rho \cos \phi \cos \phi = \rho \cos^2 \phi = \sqrt{x^2 + y^2} \cdot \left(\frac{x}{\sqrt{x^2 + y^2}} \right) \\ &= \frac{x^2}{\sqrt{x^2 + y^2}} \end{aligned}$$

$$\begin{aligned} A_y &= A \hat{a}_y \\ &= \rho \cos \phi \sin \phi = \sqrt{x^2 + y^2} \cdot \frac{x}{\sqrt{x^2 + y^2}} \cdot \frac{y}{\sqrt{x^2 + y^2}} \\ &= \frac{xy}{\sqrt{x^2 + y^2}} \end{aligned}$$

$$A_z = \sqrt{x^2 + y^2} \cdot z \cdot \frac{y}{\sqrt{x^2 + y^2}} = yz$$

$$\therefore \mathbf{A} = \frac{x^2}{\sqrt{x^2 + y^2}} \hat{a}_x + \frac{xy}{\sqrt{x^2 + y^2}} \hat{a}_y + yz \hat{a}_z$$

$$\begin{aligned} \mathbf{A}|_{(1,0,1)} &= \frac{1}{\sqrt{1}} \hat{a}_x + 0 + 0 \\ &= \hat{a}_x \end{aligned}$$

$$\therefore |\mathbf{A}|_{(1,0,1)} = 1$$

Examples: 4

A given vector function is defined by $\mathbf{F} = y \hat{a}_x + x \hat{a}_y$. Evaluate the scalar line integral from a point



P1(1, 1, -1) to P2(2, 4, -1).

- along the parabola $y = x^2$
- along the line joining the two points.

Is $\mathbf{F}$ a conservative field?

Solution:

$$\mathbf{F} = y \hat{a}_x + x \hat{a}_y$$

$$d\mathbf{l} = dx \hat{a}_x + dy \hat{a}_y$$

$$\therefore \mathbf{F} \cdot d\mathbf{l} = y dx + x dy$$

- For evaluating the line integral along the parabola $y = x^2$, we find that

$$dy = 2x dx$$

$$\begin{aligned} \therefore \int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{l} &= \int_1^2 x^2 dx + 2x^2 dx \\ &= \int_1^2 3x^2 dx = \left[x^3 \right]_1^2 = 7 \end{aligned}$$

- In this case we observe that $z_1 = z_2 = -1$, hence the line joining the points P1 and P2 lies in the $z = -1$ plane and can be represented by the equation

$$y - 1 = \frac{4 - 1}{2 - 1} (x - 1)$$

$$\text{Or, } y = 3x - 2$$

$$\therefore dy = 3 dx$$

$$\mathbf{F} \cdot d\mathbf{l} = (3x - 2)dx + x \cdot 3dx$$

$$= (6x - 2)dx$$

$$\therefore \int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{l} = \int_1^2 (6x - 2) dx$$

$$= \left[\frac{6x^2}{2} - 2x \right]_1^2$$

$$= [12 - 4 - 3 + 2]$$

$$= 7$$

The field $\mathbf{F}$ is a conservative field.

Examples: 5.



If $\mathbf{D} = \frac{1}{r} \hat{a}_r$, calculate $\int_S \mathbf{D} \cdot d\mathbf{S}$ over a hemispherical surface bounded by $r=2$ & $0 \leq \theta \leq \pi/2$.

Solution:

In spherical polar coordinates

$$d\mathbf{S} = r^2 \sin \theta d\theta d\phi \hat{a}_r$$

$$\therefore \int_S \mathbf{D} \cdot d\mathbf{S} = \int_0^2 \int_0^{\pi/2} 2 \sin \theta d\theta d\phi$$

$$= 4\pi \int_0^{\pi/2} \sin \theta d\theta$$

$$= 4\pi$$



HND II: Electrical & Electronics Engineering

Course CODE: EEC 431

Course TITLE: Electromagnetic Field Theory

Course Lecturer:

Dr. Aliyu A. B. Mayobani MSc. Ph.D. AFHEA FHEA MIET MIEEE

TOPIC 1: Principles & applications of Electrostatics



PRINCIPLES & APPLICATIONS OF ELECTROSTATICS

Coulombs' LAW

The law expresses the force between two charges Q_1 and Q_2 as shown in Fig. 1.0 below. If these charges are positive, the force is repulsive.

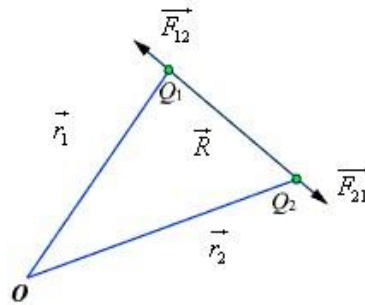


Figure 1.0: Two-point charges Q_1 and Q_2

The law states that the force between two-point charges Q_1 and Q_2 separated in vacuum or free space by a distance R which is large compared to their size... is directly proportional to the product of their charges and inversely proportional to the square of the distance between them. This force acts along the line joining the two charges. Mathematically:

$$F = \frac{kQ_1Q_2}{R^2} \quad (1)$$

Where:

Q_1 and Q_2 : point charge expressed in Coulombs(C)

F force expressed in Newtons (N)

k is the constant proportionality expressed as $k = \frac{1}{4\pi\epsilon_0}$

ϵ_0 is constant called the permittivity of free space its magnitude = $8.85 \times 10^{-12} = \frac{1}{36\pi} \times 10^{-9} \text{ F/m}$

Force on a Point Charge Place on External Field

Note that, we assumed the charges are placed in a free space. However, if the charges are any other dielectric medium, then the relative permittivity of that dielectric medium or the dielectric constant of the medium must be used.

Thus, the above equation can be rewritten as

$$\epsilon = \epsilon_0 \epsilon_r \text{ instead where } \epsilon_r$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1Q_2}{R^2} \quad (2)$$



EEC 431: LECTURE NOTES

As shown in the Fig. 1.0 above, let the position vectors of the point charges Q_1 and Q_2 are given by $\vec{r}_1$ and $\vec{r}_2$. Let $\vec{F}_{12}$ represent the force on Q_1 due to charge Q_2 . Thus, the charges

are separated by a distance of $R = |\vec{r}_1 - \vec{r}_2| = |\vec{r}_2 - \vec{r}_1|$. Then we can define the unit vectors as

$$\hat{a}_{12} = \frac{(\vec{r}_2 - \vec{r}_1)}{R} \quad \text{and} \quad \hat{a}_{21} = \frac{(\vec{r}_1 - \vec{r}_2)}{R}$$

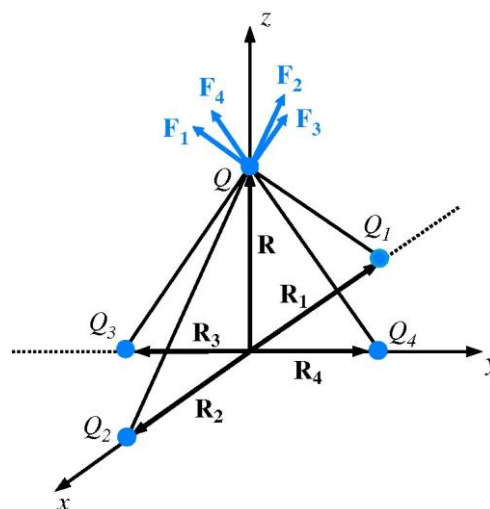
$$\vec{F}_{12} \text{ can be defined as } \vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \hat{a}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \frac{(\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|^3}$$

Numerical Example: Two identical charges are located on the x-axis at $x = 3$ and $x = 7$. At what point in space is the net electric field zero?

Solution: Since both charges are on the x-axis, the point at which the fields due to the two charges can cancel has to lie on the x-axis also. Intuitively, since the two charges are identical, that point is midway between them at (5,0,0).

Example: Four charges of $10 \mu\text{C}$ each are located in free space at points with Cartesian coordinates $(-3,0,0)$, $(3,0,0)$, $(0,-3,0)$, and $(0,3,0)$. Find the force on a $20\text{-}\mu\text{C}$ charge located at $(0,0,4)$. All distances are in meters.

Solution:





$$R_1 = -x^3$$

$$R_2 = x^3$$

$$R_3 = -y^3$$

$$R_4 = y^3$$

$$R = z^4$$

$$\mathbf{F}_1 = \frac{QQ_1}{4\pi\epsilon_0} \frac{\mathbf{R} - \mathbf{R}_1}{|\mathbf{R} - \mathbf{R}_1|^3} = \frac{QQ_1}{4\pi\epsilon_0} \frac{\hat{\mathbf{z}}4 + \hat{\mathbf{x}}3}{125} = \frac{QQ_1}{500\pi\epsilon_0} (\hat{\mathbf{z}}4 + \hat{\mathbf{x}}3)$$

$$\mathbf{F}_2 = \frac{QQ_2}{4\pi\epsilon_0} \frac{\mathbf{R} - \mathbf{R}_2}{|\mathbf{R} - \mathbf{R}_2|^3} = \frac{QQ_2}{4\pi\epsilon_0} \frac{\hat{\mathbf{z}}4 - \hat{\mathbf{x}}3}{125} = \frac{QQ_2}{500\pi\epsilon_0} (\hat{\mathbf{z}}4 - \hat{\mathbf{x}}3)$$

$$\mathbf{F}_3 = \frac{QQ_3}{4\pi\epsilon_0} \frac{\mathbf{R} - \mathbf{R}_3}{|\mathbf{R} - \mathbf{R}_3|^3} = \frac{QQ_3}{4\pi\epsilon_0} \frac{\hat{\mathbf{z}}4 + \hat{\mathbf{y}}3}{125} = \frac{QQ_3}{500\pi\epsilon_0} (\hat{\mathbf{z}}4 + \hat{\mathbf{y}}3)$$

$$\mathbf{F}_4 = \frac{QQ_4}{4\pi\epsilon_0} \frac{\mathbf{R} - \mathbf{R}_4}{|\mathbf{R} - \mathbf{R}_4|^3} = \frac{QQ_4}{4\pi\epsilon_0} \frac{\hat{\mathbf{z}}4 - \hat{\mathbf{y}}3}{125} = \frac{QQ_4}{500\pi\epsilon_0} (\hat{\mathbf{z}}4 - \hat{\mathbf{y}}3)$$

$$\begin{aligned} \mathbf{F} &= \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_4 \\ &= \frac{200 \times 10^{-12}}{500\pi\epsilon_0} (\hat{\mathbf{z}}16) = \hat{\mathbf{z}} \frac{32 \times 10^{-12}}{5\pi \times 8.85 \times 10^{-12}} = \hat{\mathbf{z}}0.23 \text{ (N)}. \end{aligned}$$

Intensity of Electric Field

The electric field intensity E (or the electric field strength) at a point is defined as the force per unit charge. This can be expressed mathematically as

$$\vec{E} = \lim_{Q \rightarrow 0} \frac{\vec{F}}{Q} \quad \text{or,} \quad \vec{E} = \frac{\vec{F}}{Q} \quad (3)$$

The electric field intensity E at a point r (observation point) due a point charge Q located at r^1 (source point) is given by:

$$\vec{E} = \frac{Q(\vec{r} - \vec{r}^1)}{4\pi\epsilon_0 |\vec{r} - \vec{r}^1|^3} \quad (4)$$

For a collection of N point charges $Q_1, Q_2, \dots, Q_N$ located at $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N$, the electric field intensity at point $\vec{r}$ is obtained as

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{Q_i(\vec{r} - \vec{r}_i)}{|\vec{r} - \vec{r}_i|^3} \quad (5)$$

The expression (5) can be modified suitably to compute the electric field due to a continuous distribution of charges.



EEC 431: LECTURE NOTES

For an elementary charge $dQ = \rho(\vec{r}') dv'$ i.e., considering this charge as point charge, we can write the field expression as:

$$d\vec{E} = \frac{dQ(\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} = \frac{\rho(\vec{r}') dv' (\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} \quad (6)$$

When this expression is integrated over the source region, we get the electric field at the point P due to this distribution of charges. Thus, the expression for the electric field at P can be written as:

$$\vec{E}(\vec{r}) = \int_V \frac{\rho(\vec{r}') (\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} dv' \quad (7)$$

Similar technique can be adopted when the charge distribution is in the form of a line charge density or a surface charge density.

$$\vec{E}(\vec{r}) = \int_L \frac{\rho_L(\vec{r}') (\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} dl' \quad (8)$$

$$\vec{E}(\vec{r}) = \int_S \frac{\rho_s(\vec{r}') (\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} dS' \quad (9)$$

Numerical Example: A square plate in the x - y plane is situated in the space defined by $-3 \text{ m} \leq x \leq 3 \text{ m}$ and $-3 \text{ m} \leq y \leq 3 \text{ m}$. Find the total charge on the plate if the surface charge density is given by $\rho_s = 4y^2 \text{ (}\mu\text{C/m}^2\text{)}$.

Solution:

$$\begin{aligned} \rho_s &= 4y^2 \\ Q &= \int_S \rho_s dS \\ &= \int_{-3}^3 \int_{-3}^3 4y^2 dx dy \\ &= \frac{4y^3 x}{3} \Big|_{-3}^3 \Big|_{-3}^3 = 432 \mu\text{C} = 0.432 \text{ (mC)}. \end{aligned}$$



Electric flux density:

As stated earlier electric field intensity or simply 'Electric field' gives the strength of the field at a particular point. The electric field depends on the material media in which the field is being considered. The flux density vector is defined to be independent of the material media (as we'll see that it relates to the charge that is producing it). For a linear isotropic medium under consideration, the flux density vector is defined as:

$$\vec{D} = \epsilon \vec{E} \quad (10)$$

We define the electric flux ψ as

$$\psi = \int_S \vec{D} \cdot d\vec{s} \quad (11)$$

Gauss's LAW

Gauss's law is one of the fundamental laws of electromagnetism and **it states that...**

the total electric flux through a closed surface is equal to the total charge enclosed by the surface.

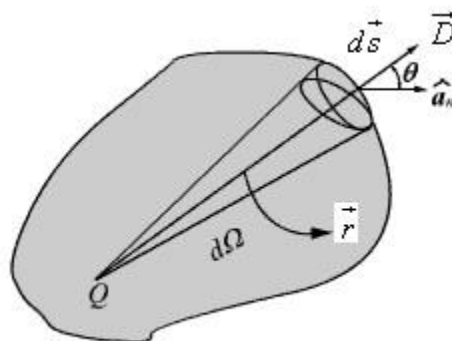


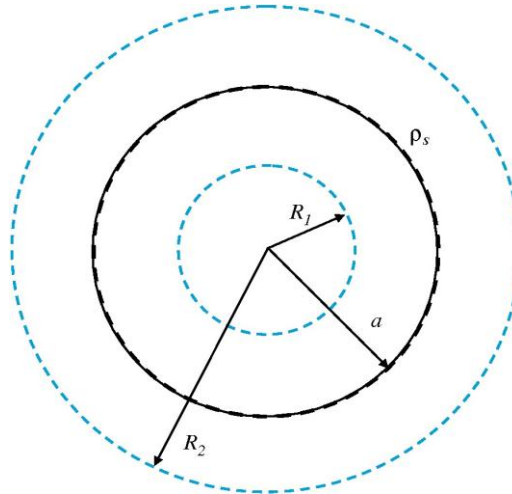
Figure 1.1: Gauss's Law

Application

Gauss's law is particularly useful in computing $\vec{E}$ or $\vec{D}$ where the charge distribution has some symmetry. illustrating the application of Gauss's Law with some examples include:

Example: A thin spherical shell of radius a carries uniform surface charge density ρ_s . Use Gauss's law to determine E .

Solution:



$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$$

Symmetry suggests that $\mathbf{D}$ is radial in direction. Hence,

$$\mathbf{D} = \hat{\mathbf{R}} D_R$$

$$d\mathbf{s} = \hat{\mathbf{R}} ds$$

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \oint_S D_R ds = D_R (4\pi R^2) = Q$$

$$D_R = \frac{Q}{4\pi R^2}$$

- For a Gaussian surface of radius $R_1 < a$, no charge is enclosed. Hence, $Q = 0$, in which case $E = 0$.
- For a Gaussian surface of radius $R_2 > a$,

$$Q = \rho_s (4\pi a^2)$$

and

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon} = \frac{\hat{\mathbf{R}}}{\epsilon} D_R = \frac{\hat{\mathbf{R}} Q}{4\pi \epsilon R_2^2} = \hat{\mathbf{R}} \frac{4\pi \rho_s a^2}{4\pi \epsilon R_2^2} = \hat{\mathbf{R}} \frac{\rho_s a^2}{\epsilon R_2^2}.$$

An infinite line charges

Let's consider the problem of determination of the electric field produced by an infinite line charge of density ρ_L C/m. Let us consider a line charge positioned along the z-axis as shown in Fig. 1.2(a). Since the line charge is assumed to be infinitely long, the electric field will be of the form as shown in Fig. 1.2(b). If we consider a close cylindrical surface as shown in Fig. 1.2(a), using Gauss's theorem we can write,

$$\rho_L l = Q = \oint_S \epsilon_0 \vec{E} \cdot d\vec{s} = \int_{S_1} \epsilon_0 \vec{E} \cdot d\vec{s} + \int_{S_2} \epsilon_0 \vec{E} \cdot d\vec{s} + \int_{S_3} \epsilon_0 \vec{E} \cdot d\vec{s} \quad (12)$$

Because the unit normal vector to areas S_1 and S_3 are perpendicular to the electric field, the surface integrals for the top and bottom surfaces evaluates to zero.

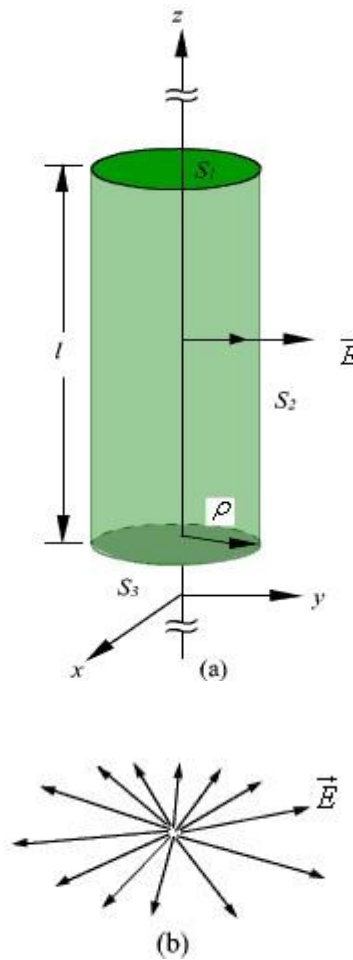


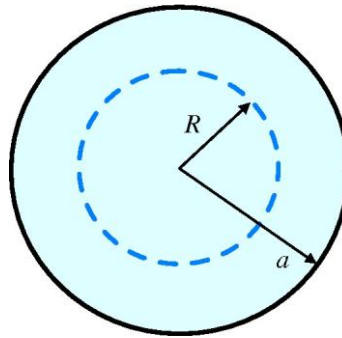
Fig 1.2: Infinite Line Charge

Hence, we can write,

$$\rho_L l = \epsilon_0 E \cdot 2\pi \rho l$$

$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 \rho} \hat{a}_\rho \quad (13)$$

Example: A spherical volume of radius a contains a uniform volume charge density ρ_v . Use Gauss's law to determine D for (a) $R \leq a$ and (b) $R \geq a$.



$$R < a$$

For $R \leq a$,

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \oint_S D_r ds = D_r (4\pi R^2)$$

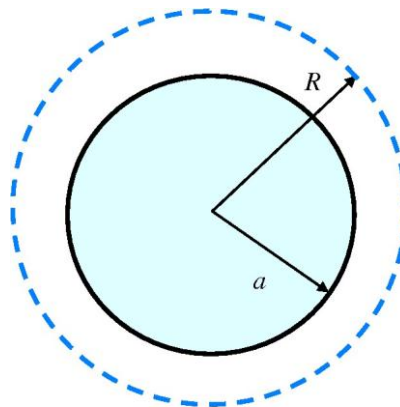
Q within a sphere of radius R is

$$Q = \frac{4}{3}\pi R^3 \rho_v$$

Hence,

$$4\pi R^2 D_R = \frac{4}{3}\pi R^3 \rho_v$$

$$D_r = \frac{\rho_v R}{3}, \quad \mathbf{D} = \hat{\mathbf{R}} D_r = \hat{\mathbf{R}} \frac{\rho_v R}{3}, \quad R \leq a.$$



$$R > a$$

For $R \geq a$, total charge in sphere is

$$Q = \frac{4}{3}\pi a^3 \rho_v,$$

$$4\pi R^2 D_R = \frac{4}{3}\pi a^3 \rho_v,$$

$$\mathbf{D} = \hat{\mathbf{R}} D_r = \hat{\mathbf{R}} \frac{\rho_v a^3}{3R^2}, \quad R \geq a.$$

Infinite Sheet of Charge

**EEC 431: LECTURE NOTES**

As a second example of application of Gauss's theorem, we consider an infinite charged sheet covering the xz plane as shown in fig. 1.3 Assuming a surface charge density of ρ_s for the infinite surface charge, if we consider a cylindrical volume having sides Δs placed symmetrically as shown in figure 2.5, we can write:

$$\oint_s \vec{D} \cdot d\vec{s} = 2D\Delta s = \rho_s \Delta s$$
$$\therefore \vec{E} = \frac{\rho_s}{2\epsilon_0} \hat{a}_y \quad (14)$$

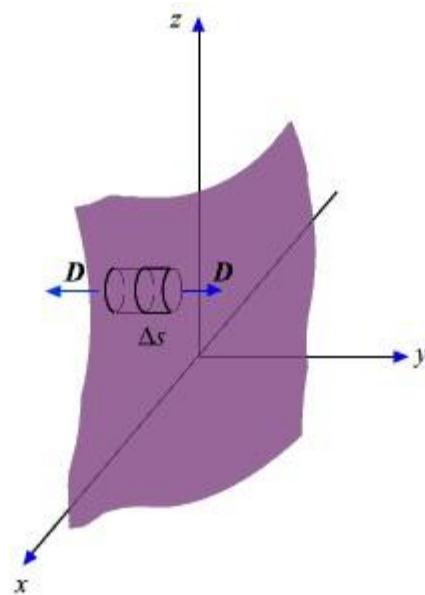


Fig 1.3: Infinite Sheet of Charge

It may be noted that the electric field strength is independent of distance. This is true for the infinite plane of charge; electric lines of force on either side of the charge will be perpendicular to the sheet and extend to infinity as parallel lines. As number of lines of force per unit area gives the strength of the field, the field becomes independent of distance. For a finite charge sheet, the field will be a function of distance.

Examples: READING ASSIGNMENT 1

As Gauss's law or Coulomb's law detailed: if the total flux through a closed surface equals the total charge enclosed by it, **Question:** What is $\vec{E}$?

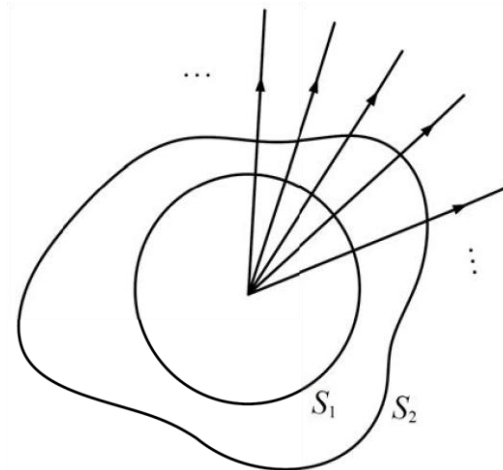


Fig. 1.4: Same amount of electric flux from a point charge passes through two surfaces S_1 and S_2 .
As detailed and shown in the Gauss's LAW diagram of fig.1.3 above, this diagram shows the same amount of electric flux from a point charge passes through two surfaces S_1 and S_2 .

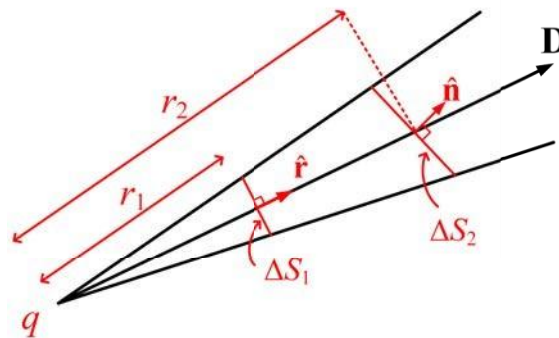


Figure 1.5: When a sliver of angular sector is taken, same amount of electric flux from a point charge passes through two incremental surfaces ΔS_1 and ΔS_2 .

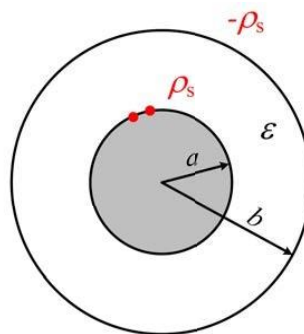


Figure 1.6: for a coaxial cylinder: Field between coaxial cylinders of unit length.

Solution Hint: Use symmetry and cylindrical coordinates as detailed in equation (12 - 13) and apply Gauss's law.

**Examples:** READING ASSIGNMENT 2

For the fields of a sphere of uniform charge density in Fig. 1.7 below. **Question:** What is $\vec{E}$?

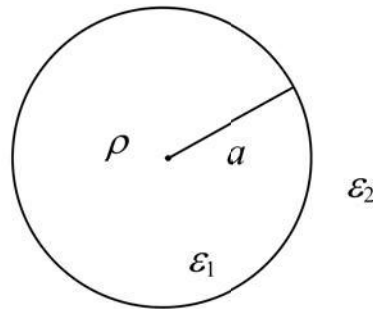


Fig. 1.7: Figure for reading assignment.

Solution Hint: Use symmetry and cylindrical coordinates as detailed in equation (12 - 13) and apply Gauss's law.



HND II: Electrical & Electronics Engineering

Course CODE: EEC 431

Course TITLE: Electromagnetic Field Theory

Course Lecturer:

Dr. Aliyu A. B. Mayobani MSc. Ph.D. AFHEA FHEA MIET MIEEE

TOPIC 2: Principles & Applications of Static Magnetic Fields



PRINCIPLES & APPLICATIONS OF STATIC MAGNETIC FIELDS

Introduction to Magnetic Fields

Electrostatic fields are generated by static charges, magnetostatic fields are generated by static currents (charges that move with constant velocity in a particular direction).

- There are several similarities between electrostatic and magnetostatic fields
- For example, as we had E and D for electrostatics, we now use B and H to examine magnetic systems
- Our study of these fields allows us to evaluate and solve for a tremendous number of electric and electromechanical devices.
- Furthermore, this study, will provide the basis for formulating an universal theory of Electromagnetic Fields that is utilized in almost every aspect of electrical engineering.

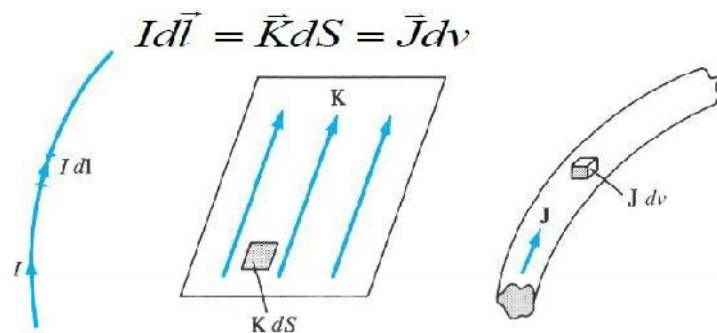
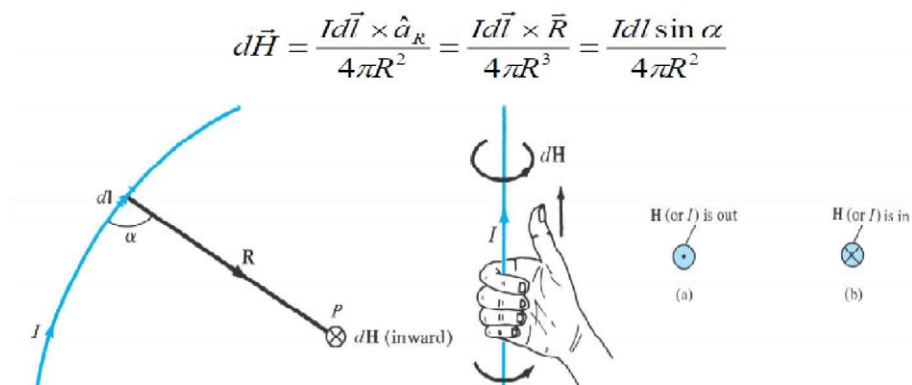
Analysis between Electric & Magnetic Fields

	Electric	Magnetic
• Basic Laws	$\vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \hat{a}_r$ $\oint \vec{D} \cdot d\vec{S} = Q_{enc}$	$d\vec{B} = \frac{\mu_0 I d\vec{l} \times \hat{a}_r}{4\pi R^2}$ $\oint \vec{H} \cdot d\vec{l} = I_{enc}$
• Force Law	$\vec{F} = Q\vec{E}$	$\vec{F} = Q\vec{u} \times \vec{B}$
• Source Element	dQ	$Q\vec{u} = Id\vec{l}$
• Field intensity	$E = \frac{V}{l} (V/m)$	$H = \frac{I}{l} (A/m)$
• Flux density	$\vec{D} = \frac{\psi}{S} (C/m^2)$	$\vec{B} = \frac{\psi}{S} (Wb/m^2)$
• Relationship Between Fields	$\vec{D} = \epsilon\vec{E}$	$\vec{B} = \mu\vec{H}$
• Potentials	$\vec{E} = -\nabla V$ $V = \int \frac{\rho_L dl}{4\pi\epsilon r}$ $\psi = \oint \vec{D} \cdot d\vec{S}$	$\vec{H} = -\nabla V_m, (\vec{J} = 0)$ $A = \int \frac{\mu I d\vec{l}}{4\pi R}$ $\psi = \oint \vec{B} \cdot d\vec{S}$
• Flux	$\psi = Q = CV$ $I = C \frac{dV}{dt}$	$\psi = LI$ $I = L \frac{d\psi}{dt}$
• Energy Density	$w_E = \frac{1}{2} \vec{D} \cdot \vec{E}$	$w_E = \frac{1}{2} \vec{B} \cdot \vec{H}$
• Poisson's Eqn.	$\nabla^2 V = -\frac{\rho_v}{\epsilon}$	$\nabla^2 A = -\mu J$



Biot-Savart's Law

The differential magnetic field intensity, $d\vec{H}$, produced at a point P , by the differential current element, $I d\vec{l}$, is proportional to the product $I d\vec{l}$ and the sine of the angle between the element and the line joining P to the element and is inversely proportional to the square of the distance, R , between P and the element



Considering different current distributions (as shown above) we can rewrite expression for field intensity as below,

$$\vec{H} = \int_L \frac{I d\vec{l} \times \hat{a}_R}{4\pi R^2}$$

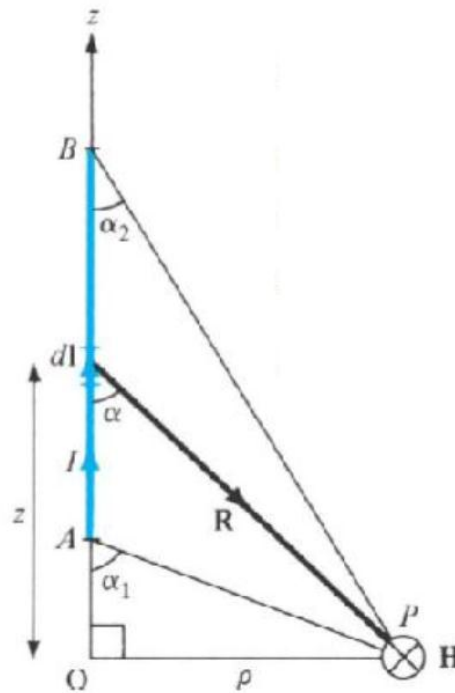
$$\vec{H} = \int_S \frac{\vec{K} dS \times \hat{a}_R}{4\pi R^2}$$

$$\vec{H} = \int_v \frac{\vec{J} dv \times \hat{a}_R}{4\pi R^2}$$



H Field From a Strait Current Carrying Filament

- The H field is determined for a strait filament of current in a manner very similar to that of the electric field determined from a line charge.



$$\vec{H} = \int_L \frac{I d\vec{l} \times \vec{R}}{4\pi R^3}$$

$$d\vec{l} = dz \hat{a}_z$$

$$\vec{R} = \rho \hat{a}_\rho - z \hat{a}_z$$

$$d\vec{l} \times \vec{R} = \rho dz \hat{a}_\phi$$

$$\vec{H} = \int_L \frac{I \rho dz \hat{a}_\phi}{4\pi (\rho^2 + z^2)^{3/2}}$$

Now,



$$\begin{aligned}z &= \rho \cot \alpha \\dz &= -\rho (\csc^2 \alpha) d\alpha \\ \vec{H} &= \int_L -\frac{I \rho^2 (\csc^2 \alpha) d\alpha \hat{a}_\phi}{4\pi (\rho \csc \alpha)^3} \\ \vec{H} &= -\frac{I}{4\pi \rho} \int_{\alpha_1}^{\alpha_2} (\sin \alpha) d\alpha \hat{a}_\phi \\ \vec{H} &= \frac{I}{4\pi \rho} (\cos \alpha_2 - \cos \alpha_1) \hat{a}_\phi \\ \vec{H} &= \frac{I}{4\pi \rho} \hat{a}_\phi \quad \text{Line from } z = 0 \text{ to } \infty \\ \vec{H} &= \frac{I}{2\pi \rho} \hat{a}_\phi \quad \text{Line from } z = -\infty \text{ to } \infty\end{aligned}$$

Using Biot-savart's law we can find the expression for field intensity due to different current carrying conductor configurations.

Ampere's law:

The line integral of $\vec{H}$ around a closed path is the same as the net current, I_{enc} , enclosed by the path,

$$\oint \vec{H} \cdot d\vec{l} = I_{enc}$$

- Similar to Gauss' law since Ampere's law is easily used to determine H when the current distribution is symmetrical.
- Ampere's law ALWAYS holds, even if the current distribution is NOT symmetrical, however the equation is typically used for symmetric cases.
- Like Gauss and Coulomb's Laws, Ampere's law is a special case of the Biot-Savart law and can be derived directly from it.

**Derivation of Divergence's and Stoke's Theorem:**

Recalled from the VECTOR introduction class the details on the following:

Divergence of a Vector:

- The divergence of a vector, $\mathbf{A}$, at any given point P is the outward flux per unit volume as volume shrinks about P.

$$\text{div} \vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

Divergence Theorem:

- The divergence theorem states that the total outward flux of a vector field, $\mathbf{A}$, through the closed surface, S, is the same as the volume integral of the divergence of $\mathbf{A}$.
- This theorem is easily shown from the equation for the divergence of a vector field.

$$\begin{aligned} \vec{A} &= A_1 \hat{a}_1 + A_2 \hat{a}_2 + A_3 \hat{a}_3 \\ \text{div} \vec{A} &= \nabla \cdot \vec{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v} \\ \int_v \nabla \cdot \vec{A} dv &= \oint_s \vec{A} \cdot d\vec{S} \end{aligned}$$

Stokes Theorem:

Stokes theorem states that the circulation of a vector field $\mathbf{A}$, around a closed path, L is equal to the surface integral of the curl of $\mathbf{A}$ over the open surface S bounded by L. This theorem has been proven to hold as long as $\mathbf{A}$ and the curl of $\mathbf{A}$ are continuous along the closed surface S of a closed path L

- This theorem is easily shown from the equation for the curl of a vector field.

$$\begin{aligned} \vec{A} &= A_1 \hat{a}_1 + A_2 \hat{a}_2 + A_3 \hat{a}_3 \\ \text{curl} \vec{A} &= \nabla \times \vec{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \vec{A} \cdot d\vec{l}}{\Delta S} \right) \hat{a}_n \\ \oint_L \vec{A} \cdot d\vec{l} &= \oint_s (\nabla \times \vec{A}) \cdot d\vec{S} \end{aligned}$$



Applying Stokes's theorem on Ampere's law

Applying Stokes's theorem on Ampere's law, we have,

$$I_{enc} = \oint_L \vec{H} \cdot d\vec{l} = \int_S (\nabla \times \vec{H}) \cdot d\vec{S}$$

We can also write using current density,

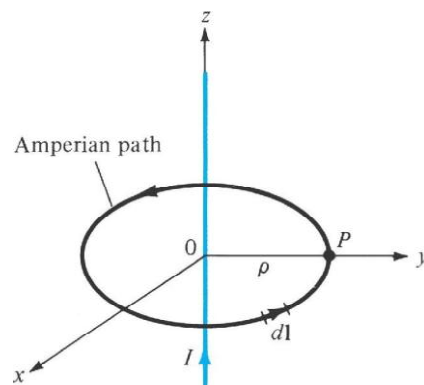
$$I_{enc} = \int_S \vec{J} \cdot d\vec{S}$$

So, from the above two equations,

$$\nabla \times \vec{H} = \vec{J}$$

Applications of Ampere's Circuit Law

- A simple application of Ampere's law can be used to easily derive the magnetic field intensity from an infinite line current ,



$$I_{enc} = \oint \vec{H} \cdot d\vec{l}$$

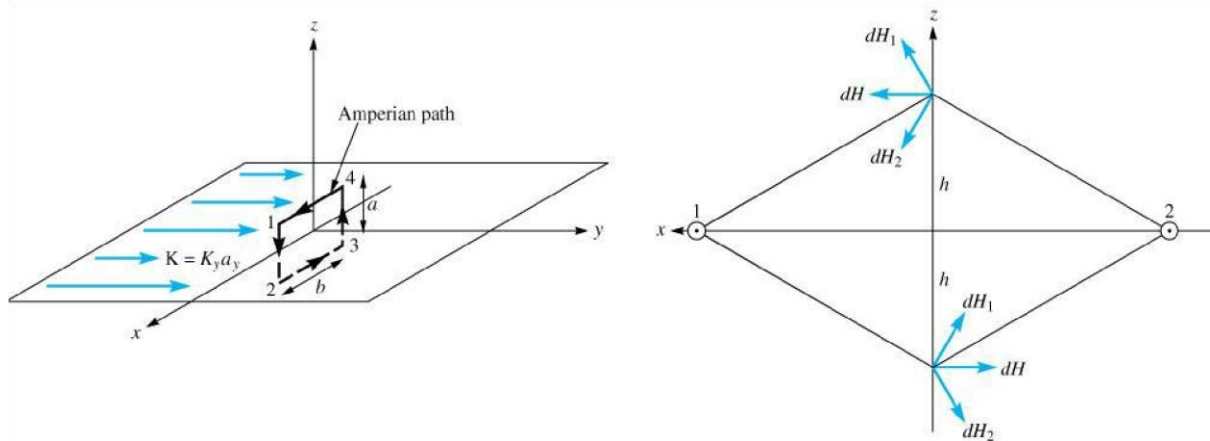
$$I = \int H_{\phi} \hat{a}_{\phi} \cdot \rho d\phi \hat{a}_{\phi} = H_{\phi} \int \rho d\phi = H_{\phi} \cdot 2\pi\rho$$

$$H_{\phi} = \frac{I}{2\pi\rho} \hat{a}_{\phi}$$



Ampere's Circuit Law: Infinite Sheet of Current

- Consider an infinite sheet of current in the $z=0$ plane with a uniform current density, $\mathbf{K} = K_y \mathbf{a}_y$.



$$I_{enc} = \oint \vec{H} \cdot d\vec{l} = K_y b \quad \text{Apply Ampere's law}$$

$$\vec{H} = \begin{cases} H_0 \hat{a}_x, & z > 0 \\ -H_0 \hat{a}_x, & z < 0 \end{cases}$$

$$\oint \vec{H} \cdot d\vec{l} = \left(\int_1^2 + \int_2^3 + \int_3^4 + \int_4^1 \right) \vec{H} \cdot d\vec{l}$$

$$= 0(-a) - H_0(-b) + 0(a) + H_0(b) = 2H_0 b$$

from Ampere's law and integral summation

$$H_0 = K_y$$

$$\vec{H} = \begin{cases} \frac{1}{2} K_y \hat{a}_x, & z > 0 \\ -\frac{1}{2} K_y \hat{a}_x, & z < 0 \end{cases}$$

Thus for an infinite sheet of charge,

$$\vec{H} = \frac{1}{2} \vec{K} \times \hat{a}_n$$

So, by using Ampere's circuital law, the expressions for magnetic field intensity of different structures can be derived.

Magnetic Flux Density

- Magnetic Flux density, $\mathbf{B}$, is the magnetic equivalent of the electric flux density, $\mathbf{D}$. As such, one can define,

$$\vec{B} = \mu_0 \vec{H}$$



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where $\mu_0 = 4\pi \times 10^{-7} \text{ H / m}$

- Similarly, Ampere's Law is,

$$I_{enc} = \oint \frac{\vec{B}}{\mu_0} \cdot d\vec{l}$$

- And the Magnetic flux through a surface is,

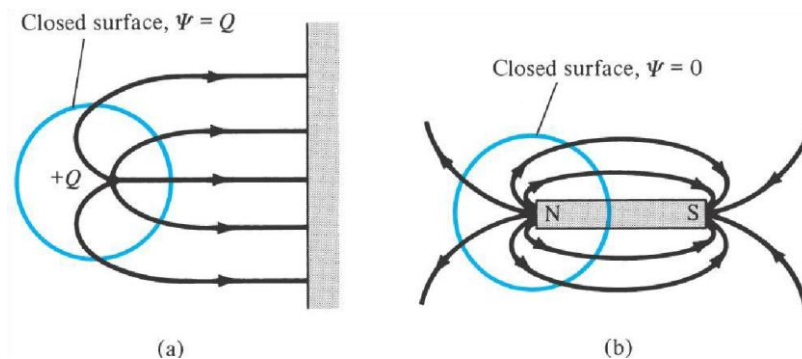
$$\psi = \int_S \vec{B} \cdot d\vec{S} = \mu_0 \int_S \vec{H} \cdot d\vec{S}$$

- The magnetic flux through an enclosed system is,

$$\psi = \int_S \vec{B} \cdot d\vec{S} = \int_V (\nabla \cdot \vec{B}) dV$$

- Unlike electrostatic flux however, magnetic flux always follows a closed path and fold in on themselves. This simple statement has profound consequences. In electrostatics, we can easily define a point charge in which electric fields emanate to infinity.

However, the solenoidal nature of the magnetic field requires magnetic flux to travel from a positive (north) to a negative (south) pole and it is not possible to have a single magnetic pole at any time.



–There are NO magnetic monopoles, stipulating that an isolated magnetic charge DOES NOT EXIST

–The minimum field requirement for magnetics is a dipole. So, mathematically,

$$\nabla \cdot \vec{B} = 0$$

**Maxwell's Eqns. for Static Fields**

Differential Form	Integral Form	Remarks
$\nabla \cdot \vec{D} = \rho_v$	$\oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_v dv$	Gauss's Law
$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot d\vec{S} = 0$	Nonexistence of the Magnetic Monopole
$\nabla \times \vec{E} = 0$	$\oint_L \vec{E} \cdot d\vec{l} = 0$	Conservative nature of the Electric Field
$\nabla \times \vec{H} = \vec{J}$	$\oint_L \vec{H} \cdot d\vec{l} = \oint_S \vec{J} \cdot d\vec{S}$	Ampere's Law

Magnetic (Scalar & Vector) Potential

We can define a magnetic field using the following requirements.

$$\begin{aligned}\nabla \times \nabla V &= 0 \\ \nabla \cdot (\nabla \times \vec{A}) &= 0\end{aligned}$$

- Just as $\vec{E} = -\nabla V$, we can define a magnetic scalar potential V_m related to $\vec{H}$ when the current density is zero as

$$\begin{aligned}\vec{H} &= -\nabla V_m, \vec{J} = 0 \\ \vec{J} &= \nabla \times \vec{H} = \nabla \times (-\nabla V_m) = 0 \\ \nabla^2 V_m &= 0, \vec{J} = 0\end{aligned}$$

- The requirement for a solenoidal field (and Maxwell's 4th law of electrostatics) stipulates

$$\nabla \cdot \vec{B} = 0$$

- And we can therefore define a magnetic vector potential, $\vec{A}$, as

$$\vec{B} = \nabla \times \vec{A}$$

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- Just as we defined the Electric Potential as $V = \int \frac{dQ}{4\pi\epsilon_0 r^2}$, We can define the Magnetic Vector Potential

as,

$$\begin{aligned}\vec{A} &= \int_L \frac{\mu_0 I d\vec{l}}{4\pi R} && \text{for Line Current} \\ \vec{A} &= \int_L \frac{\mu_0 \vec{K} dS}{4\pi R} && \text{for Surface Current} \\ \vec{A} &= \int_L \frac{\mu_0 \vec{J} dv}{4\pi R} && \text{for Volume Current}\end{aligned}$$

Forces Magnetic Field**Lorentz Force Law:**

- The force on a charged particle in an electric field is simply $\vec{F} = q\vec{E}$
- However, in the presence of an electromagnetic field an additional force is imposed from the charge displacement of velocity, $\vec{u}$, quantified by the magnetic field, $\vec{B}$.
- The combined force is defined by Lorentz Force Law:

$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$$

- Equating the Lorentz force to Newton's force equation, we have,

$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B}) = m\vec{a} = m \frac{d\vec{u}}{dt}$$

Where 'a' is the acceleration of the particle is space.

Force on a Current Element

- One can also use field calculations to determine the force acting on a current element, $I d\vec{l} = \vec{K} dS = \vec{J} dv$, due to an applied external magnetic field $\vec{B}$.

Assume that a copper wire carries a current density, $\vec{J} = \rho_v \vec{u}$.

- We know:

$$I d\vec{l} = \vec{J} dv = \rho_v \vec{u} dv = dQ \vec{u}$$

- And that:

$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$$

$$\text{where } \vec{E} \rightarrow 0$$

$$\vec{F} = q(\vec{u} \times \vec{B})$$



Thus, we can solve for the force on the first wire:

$$d\vec{F} = dq(\vec{u} \times \vec{B}) = Id\vec{l} \times \vec{B}$$

$$\vec{F} = \oint_L Id\vec{l} \times \vec{B}$$

Likewise

$$\vec{F} = \oint_L \vec{K} dS \times \vec{B} = \vec{F} = \oint_L \vec{J} dv \times \vec{B}$$

Examples: 1

An electron moving in the positive x-direction perpendicular to a magnetic field experiences a deflection in the negative z-direction. What is the direction of the magnetic field?

Solution: The magnetic force acting on a moving charged particle is

$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$$

$$q = -e$$

$$\mathbf{u} = \hat{\mathbf{x}}u$$

$$\mathbf{F}_m = -\hat{\mathbf{z}}F_m$$

$$-\hat{\mathbf{z}}F_m = -\hat{\mathbf{x}}ue \times \mathbf{B}$$

For the cross product to apply, B has to be in the positive y-direction.

Examples: 2

A proton moving with a speed of 2×10^6 m/s through a magnetic field with magnetic flux density of 2.5 T experiences a magnetic force of magnitude 4×10^{-13} N.

What is the angle between the magnetic field and the proton's velocity?

Solution:

$$F = quB \sin \theta$$

$$\sin \theta = \frac{F}{quB}$$

$$\begin{aligned} &= \frac{4 \times 10^{-13}}{1.6 \times 10^{-19} \times 2 \times 10^6 \times 2.5} \\ &= 0.5 \end{aligned}$$

$$\theta = \sin^{-1} 0.5 = 30^\circ \text{ or } 150^\circ.$$

**Examples: 3**

A charged particle with velocity $\mathbf{u}$ is moving in a medium containing uniform fields $\mathbf{E} = x\hat{x}$ and $\mathbf{B} = y\hat{y}$.

What should $\mathbf{u}$ be so that the particle experiences no net force on it?

Solution:

$$\mathbf{F}_e = q\mathbf{E} = xq\hat{x}$$
$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B} = q(\mathbf{u} \times y\hat{y})$$

For net force to be zero, $\mathbf{F}_m$ has to be along $-x\hat{x}$, which requires $\mathbf{u}$ to be along $+z\hat{z}$. Thus,

$$qE = quB$$
$$u = \frac{E}{B}$$
$$\mathbf{u} = \hat{z} \frac{E}{B}.$$

If $\mathbf{u}$ also has a y -component, that component will exercise no force on the particle.

Examples: 4

A horizontal wire with a mass per unit length of 0.2 kg/m carries a current of 4 A in the $+x$ -direction. If the wire is placed in a uniform magnetic flux density $\mathbf{B}$, what should the direction and minimum magnitude of $\mathbf{B}$ be in order to magnetically lift the wire vertically upward? (Hint: The acceleration due to gravity is $g = -z\hat{z}9.8 \text{ m/s}^2$.)

Solution: For a length l ,

$$\mathbf{F}_g = -\hat{z}0.2l \times 9.8 = -\hat{z}1.96l \quad (\text{N})$$
$$\mathbf{F}_m = \hat{x}Il \times \mathbf{B}$$

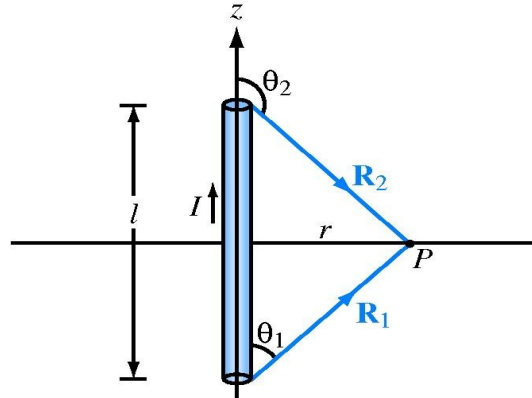
For $\mathbf{F}_m + \mathbf{F}_g = 0$, $\mathbf{F}_m$ has to be along $+z\hat{z}$, which means that $\mathbf{B}$ has to be along $+y\hat{y}$. Hence,

$$1.96l = IlB$$
$$B = \frac{1.96}{I} = 0.49 \text{ (T), and}$$
$$\mathbf{B} = \hat{y}0.49 \text{ (T).}$$

Examples: 5

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A semi-infinite linear conductor extends between $z = 0$ and $z = \infty$ along the z -axis. If the current I in the conductor flows along the positive z -direction, find H at a point in the x - y plane at a radial distance r from the conductor.



$$\mathbf{H} = \hat{\phi} \frac{I}{4\pi r} (\cos \theta_1 - \cos \theta_2)$$

For a conductor extending from $z = 0$ to $z = \infty$, $\theta_1 = 0$ and $\theta_2 = \pi$. Hence,

$$\mathbf{H} = \hat{\phi} \frac{I}{4\pi r} (1 + 1) = \hat{\phi} \frac{I}{2\pi r} \quad (\text{A/m}).$$



HND II: Electrical & Electronics Engineering

Course CODE: EEC 431

Course TITLE: Electromagnetic Field Theory

Course Lecturer:

Dr. Aliyu A. B. Mayobani MSc. Ph.D. AFHEA FHEA MIET MIEEE

TOPIC 3: Principles & Applications of Time Varying Electromagnetic Fields



PRINCIPLES & APPLICATIONS OF TIME VARYING ELECTROMAGNETIC FIELDS

Inductors and Inductance:

- We now know that closed magnetic circuit carrying current I produces a magnetic field with flux

$$\Psi = \int \vec{B} \cdot d\vec{S}$$

- We define the flux linkage between a circuit with N identical turns as,

$$\lambda = N\Psi$$

- As long as the medium the flux passes through is linear (isotropic) then the flux linkage is proportional to the current I producing it and can be written as,

$$\lambda = LI$$

Where L is a constant of proportionality called the inductance of the circuit. A circuit that contains inductance is said to be an inductor.

- One can equate the inductance to the magnetic flux of the circuit as

$$L = \frac{\lambda}{I} = \frac{N\Psi}{I}$$

where L is measured in units of Henrys (H) = Wb/A.

- The magnetic energy (in Joules) stored by the inductor is expressed as,

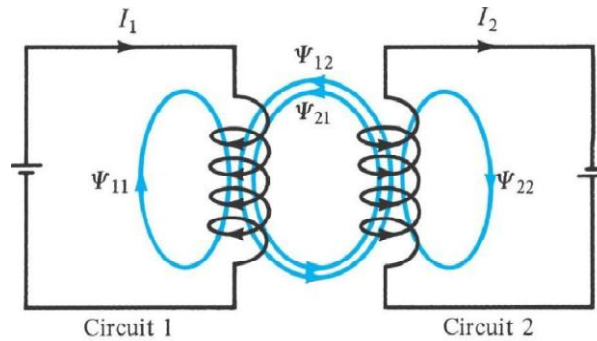
$$W_m = \frac{1}{2} LI^2$$

Inductors and Inductance:

- Since we know that magnetic fields produce forces on nearby current elements, and that those magnetic fields can be generated by an isolated or coupled set of current carrying circuits, then it is only reasonable that such circuits may induce fields and magnetization between them.



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- We can calculate the individual flux linkage between the two components as

$$\Psi_{12} = \int_{S_1} \vec{B}_2 \cdot d\vec{S}$$

- Likewise we can determine a mutual inductance between the circuits that is equal from circuit 12 as it is from circuit 21 as

$$M_{12} = \frac{\lambda_{12}}{I_2} = \frac{N_1 \Psi_{12}}{I_2}$$

- Individual inductances are

$$L_1 = \frac{\lambda_{11}}{I_1} = \frac{N_1 \Psi_1}{I_1} \quad L_2 = \frac{\lambda_{22}}{I_2} = \frac{N_2 \Psi_2}{I_2}$$

- The total magnetic energy in the circuit is

$$W_m = W_1 + W_2 + W_{12} = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 \pm M_{12} I_1 I_2$$

- Mutual inductance may be calculated by the following method,
 - Determine the internal inductance, L_{in} for the flux generated by the first inductor
 - Determine the external inductance, L_{ext} produced by the flux external of the first inductor
 - The sum of the internal and external inductance equals the individual inductances plus the mutual inductance between the elements

$$M_{12} = \frac{\lambda_{12}}{I_2} = \frac{N_1 \Psi_{12}}{I_2} \quad \Psi_{12} = \int_{S_1} \vec{B}_2 \cdot d\vec{S}$$

Magnetic Energy

We can derive a similar term as derived for electric energy, for magnetic energy using the relation for energy as a function of inductance.



$$W_m = \frac{1}{2} LI^2$$

$$\Delta L = \frac{\Delta \Psi}{\Delta I} = \frac{\mu H \Delta x \Delta y}{\Delta I}$$

$$\text{where, } \Delta I = H \Delta z$$

$$\Delta L = \frac{\mu H \Delta x \Delta y}{\Delta I} = \frac{\mu \Delta x \Delta y}{\Delta z}$$

$$\Delta W_m = \frac{1}{2} \Delta L \Delta I^2 = \frac{1}{2} \mu H^2 \Delta x \Delta y \Delta z$$

$$\Delta W_m = \frac{1}{2} \mu H^2 \Delta v$$

$$w_m = \lim_{\Delta v \rightarrow 0} \frac{\Delta W_m}{\Delta v} = \frac{1}{2} \mu H^2 = \frac{1}{2} (\vec{H} \cdot \vec{B}) = \frac{B^2}{2\mu}$$

$$W_m = \int w_m dv = \int \frac{1}{2} (\vec{H} \cdot \vec{B}) dv = \int \frac{1}{2} \mu H^2 dv$$

Magnetic Circuits

•The following relations allow one to solve magnetic field problems in a manner similar to that of electronic circuits. It provides a clear means of designing transformers, motors, generators, and relays using a lumped circuit model. The analogy between electronic and magnetic circuits is provided below.

Table 8.4 Analogy between Electric and Magnetic Circuits

Electric	Magnetic
Conductivity σ	Permeability μ
Field intensity E	Field intensity H
Current $I = \int \mathbf{J} \cdot d\mathbf{S}$	Magnetic flux $\Psi = \int \mathbf{B} \cdot d\mathbf{S}$
Current density $\mathbf{J} = \frac{\mathbf{I}}{S} = \sigma \mathbf{E}$	Flux density $\mathbf{B} = \frac{\Psi}{S} = \mu \mathbf{H}$
Electromotive force (emf) V	Magnetomotive force (mmf) $\mathcal{F}$
Resistance R	Reluctance $\mathcal{R}$
Conductance $G = \frac{1}{R}$	Permeance $\mathcal{P} = \frac{1}{\mathcal{R}}$
Ohm's law $R = \frac{V}{I} = \frac{\ell}{\sigma S}$ or $V = E\ell = IR$	Ohm's law $\mathcal{R} = \frac{\mathcal{F}}{\Psi} = \frac{\ell}{\mu S}$ or $\mathcal{F} = H\ell = \Psi \mathcal{R} = NI$
Kirchhoff's laws: $\sum I = 0$ $\sum V - \sum RI = 0$	Kirchhoff's laws: $\sum \Psi = 0$ $\sum \mathcal{F} - \sum \mathcal{R} \Psi = 0$



Faraday's Law for induced emf:

Faraday's law state that... the Induced electromotive force (emf) (in volts) in any closed circuit is equal to the time rate of change of magnetic flux by the circuit,

$$V_{emf} = -\frac{d\lambda}{dt} = -N \frac{d\Psi}{dt}$$

Where: λ is the flux linkage,

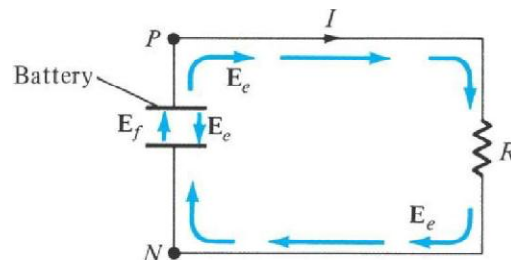
Ψ is the magnetic flux,

N is the number of turns in the inductor, and

t represents a time interval.

The negative sign shows that the induced voltage acts to oppose the flux producing it.

- The statement in blue above is known as Lenz's Law: the induced voltage acts to oppose the flux producing it.
- Examples of emf generated electric fields: electric generators, batteries, thermocouples, fuel cells, photovoltaic cells, transformers.
- To elaborate on emf, let's consider a battery circuit.



- The electrochemical action within the battery results and in emf produced electric field, $\vec{E}_f$
- Accumulated charges at the terminals provide an electrostatic field $\vec{E}_e$ that also exist that counteracts the emf generated potential.

$$\vec{E} = \vec{E}_f + \vec{E}_e$$

$$\oint_L \vec{E} \cdot d\vec{l} = \oint_L \vec{E}_f \cdot d\vec{l} + 0 = \int_N^P \vec{E}_f \cdot d\vec{l}$$

- The total emf generated in the between the two open terminals in the battery is therefore,



$$V_{emf} = \int_N^P \vec{E}_f \cdot d\vec{l} = - \int_N^P \vec{E}_e \cdot d\vec{l} = IR$$

Transformer and Motional Electromotive Forces:

- The variation of flux with time may be caused by three ways
 1. Having a stationary loop in a time-varying **B** field
 2. Having a time-varying loop in a static **B** field
 3. Having a time-varying loop in a time-varying **B** field

- A stationary loop in a time-varying **B** field

$$\nabla \times \vec{E} = - \frac{d\vec{B}}{dt}$$

- A time-varying loop in a static **B** field

$$\nabla \times \vec{E}_m = \nabla \times (\vec{u} \times \vec{B})$$

- A time-varying loop in a time-varying **B** field

$$\nabla \times \vec{E}_m = - \frac{d\vec{B}}{dt} + \nabla \times (\vec{u} \times \vec{B})$$

Displacement Current:

- Lets now examine time dependent fields from the perspective on Ampere's Law.

$$\nabla \times \vec{H} = \vec{J}$$

$$\nabla \cdot (\nabla \times \vec{H}) = 0 = \nabla \cdot \vec{J}$$

$$\nabla \cdot \vec{J} = - \frac{\partial \rho_v}{\partial t} \neq 0$$

This vector identity for the cross product is mathematically valid. However, it requires that the continuity eqn. equals zero, which is not valid from an electrostatics standpoint!

$$\nabla \times \vec{H} = \vec{J} + \vec{J}_d$$

$$\nabla \cdot (\nabla \times \vec{H}) = 0 = \nabla \cdot \vec{J} + \nabla \cdot \vec{J}_d$$

Thus, lets add an additional current density term to balance the electrostatic field requirement

$$\nabla \cdot \vec{J}_d = - \nabla \cdot \vec{J} = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t} (\nabla \cdot \vec{D}) = \nabla \cdot \frac{\partial \vec{D}}{\partial t}$$

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

We can now define the displacement current density as the time derivative of the displacement vector

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

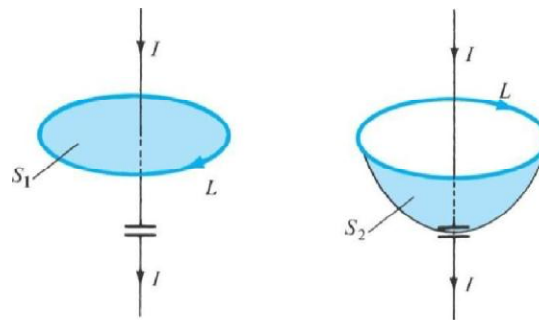
Another of Maxwell's for time varying fields

This one relates Magnetic Field Intensity to conduction and displacement current densities



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- We can apply the displacement current concept on the simple case of a capacitive element in a simple electronic circuit, as shown below.



Based on the equation for displacement current density, we can define the displacement current in a circuit as shown. Applying Ampere's circuit law to a closed path provides the following eqn. for current on the first side of the capacitive element. However, surface 2 is the opposite side of the capacitor and has no conduction current allowing for no enclosed current at surface 2. If $J = 0$ on the second surface, then J_d must be generated on the second surface to create a time displaced current equal to current on surface 1.

$$\begin{aligned}\nabla \times \vec{H} &= \vec{J} + \frac{\partial \vec{D}}{\partial t} \\ I_d &= \int \vec{J} \cdot d\vec{S} = \int \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S} \\ \oint_L \vec{H} \cdot d\vec{l} &= \int_{S_1} \vec{J} \cdot d\vec{S} = I_{enc} = I \\ \oint_L \vec{H} \cdot d\vec{l} &= \int_{S_2} \vec{J} \cdot d\vec{S} = I_{enc} = 0\end{aligned}$$

If $J = 0$ on the second surface, then J_d must be generated on the second surface to create a time displaced current equal to current on surface 1.

$$\oint_L \vec{H} \cdot d\vec{l} = \int_{S_2} \vec{J}_d \cdot d\vec{S} = \frac{\partial}{\partial t} \int_{S_2} \vec{D} \cdot d\vec{S} = \frac{dQ}{dt} = I$$

We know,

$$D = \epsilon E = \epsilon \frac{V}{d}$$

So,



$$J_d = \frac{\partial D}{\partial t} = \frac{\epsilon}{d} \frac{dV}{dt}$$

$$\Rightarrow I_d = \bar{J}_d \cdot \bar{S} = \frac{\epsilon S}{d} \frac{dV}{dt} = C \frac{dV}{dt}$$

from _surface_1

$$I_c = \frac{dQ}{dt} = S \frac{d\rho_s}{dt} = S \frac{dD}{dt} = \epsilon S \frac{dE}{dt} = \frac{\epsilon S}{d} \frac{dV}{dt} = C \frac{dV}{dt}$$

Maxwell's Time Dependent Equations

The Maxwell's equations for time dependent fields are,

Differential Form	Integral Form	Remarks
$\nabla \cdot \vec{D} = \rho_v$	$\oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_v dv$	Gauss's Law
$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot d\vec{S} = 0$	Nonexistence of the Magnetic Monopole
$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint_L \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{S}$	Faraday's Law
$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\oint_L \vec{H} \cdot d\vec{l} = \oint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{S}$	Ampere's Circuit Law

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HND II: Electrical & Electronics Engineering

Course CODE: EEC 431

Course TITLE: Electromagnetic Field Theory

Course Lecturer:

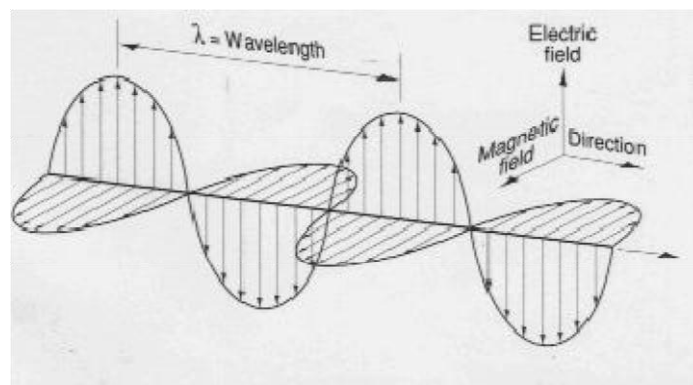
Dr. Aliyu A. B. Mayobani MSc. Ph.D. AFHEA FHEA MIET MIEEE

TOPIC 4: Principles & Applications of Plane Waves

PRINCIPLES & APPLICATIONS OF PLANE WAVES

Plane Wave:

A uniform plane wave is the wave that the electric field, E or magnetic field, H in same direction, same magnitude and same phase in infinite planes perpendicular to the direction of propagation. A plane wave has no electric field, and magnetic field, components along its direction of propagation.



Wave Equations:

If the wave is in simple (linear, isotropic and homogeneous) nonconducting medium ($\sigma=0$), Maxwell's equation reduce to,

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{H} = 0$$



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The first-order differential equations in the two variables E and H . They can combine to give E or H alone using second-order equation.

Using Maxwell's equation,

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \quad (1) \quad \vec{\nabla} \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t} \quad (2) \quad \vec{\nabla} \cdot \vec{E} = 0 \quad (3)$$

The curl of equation of (1)

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\mu \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{H})$$

Replacing in equation (2)

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

We know that $\vec{\nabla} \times \vec{\nabla} \times \vec{E} = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E}$ because of equation (3), thus the wave equation is

$$\vec{\nabla}^2 \vec{E} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

The wave equation also can written as

$$\vec{\nabla}^2 \vec{E} - k^2 \vec{E} = 0 \quad \text{-----(a)}$$

Assuming an implicit time dependence $e^{j\omega t}$ in the field vector. Equation (a) also called Helmholtz equation. The k is called the wave number or propagation constant.

$$k = k_o \sqrt{\epsilon_r} = \frac{2\pi f}{c} \sqrt{\epsilon_r} \quad \text{and} \quad c = \frac{1}{\sqrt{\epsilon \mu}}$$

where c is the velocity of light in free space.

For magnetic intensity domain, H , we have,

$$\vec{\nabla}^2 \vec{H} - \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2} = 0 \quad \text{or} \quad \vec{\nabla}^2 \vec{H} - \mu_r \epsilon_r k_o^2 \vec{H} = 0$$

For a uniform plane wave with an electric field $E = x E_x$



traveling in the z-direction, the wave equation can be reduced as

$$\frac{\partial^2 \vec{E}_x(z)}{\partial z^2} - k^2 \vec{E}_x(z) = 0$$

The solution of this wave equation,

$$\begin{aligned}\vec{E}(z) &= \hat{x}E_x \\ &= \hat{x}E_o e^{-kz} \\ &= \hat{x}E_o e^{-\alpha z} e^{-j\beta z}\end{aligned}$$

Where α is the attenuation constant of the medium and β is its phase constant.

The associated magnetic field, H ,

$$\begin{aligned}\vec{H}(z) &= \hat{y}H_y \\ &= \hat{y} \frac{\vec{E}_x}{\eta} \\ &= \hat{y} \frac{E_o}{\eta} e^{-\alpha z} e^{-j\beta z}\end{aligned}$$

where η is the intrinsic impedance of the medium.

The k is called the wave number or propagation constant.

$$\begin{aligned}k^2 &= k_o^2 \epsilon_r \mu_r \\ k^2 &= k_o^2 \mu_r (\epsilon_r' - j\epsilon_r'')\end{aligned}$$

The wave number can also be written in terms of α and β .

$$\begin{aligned}k^2 &= (\alpha + j\beta)^2 \\ &= (\alpha^2 - \beta^2) + j2\alpha\beta\end{aligned}$$

Thus,

$$\alpha^2 - \beta^2 = k_o^2 \mu_r \epsilon_r' \quad (1)$$

$$2\alpha\beta = -k_o^2 \mu_r \epsilon_r'' \quad (2)$$



By solving (1) & (2),

$$\alpha = \sqrt{\frac{k_o^2 \mu_r \epsilon_r'}{2} \left(\sqrt{1 + \left(\frac{\epsilon_r''}{\epsilon_r'} \right)^2} - 1 \right)}$$

$$\beta = \sqrt{\frac{k_o^2 \mu_r \epsilon_r'}{2} \left(\sqrt{1 + \left(\frac{\epsilon_r''}{\epsilon_r'} \right)^2} + 1 \right)}$$

So, for different medium,

Lossless Medium	Low-loss Medium	Conductor
$(\sigma = 0)$	$(\epsilon''/\epsilon' \neq 0)$	$(\epsilon''/\epsilon' \gg \infty)$
$\alpha = 0$	$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$	$\alpha = \sqrt{\pi f \mu \sigma}$
$\beta = \omega \sqrt{\mu \epsilon}$	$\beta = \omega \sqrt{\mu \epsilon}$	$\beta = \sqrt{\pi f \mu \sigma}$

Electromagnetic Phenomena are described by using four Maxwell's equations:



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Maxwell's equation			
Gauss's Law (Electric fields)	Integral form:	Description	Information
	$\underbrace{\epsilon_0 \oint \vec{E} \cdot d\vec{S}}_{\text{Left}} = \underbrace{q}_{\text{Right}}$	Left side: The number of electric field lines – perpendicularly passing through to a closed surface, $\vec{S}$ Right side: Total amount of charge, q contained within that surface, .	Electric charge produces an electric field, $\vec{E}$ and the flux of that field passing through any closed surface is proportional to the total charge, q contained within that surface. Charge on an insulated conductor moves outward surface.
	Differential form:	Description	Information
	$\underbrace{\epsilon_0 \vec{\nabla} \cdot \vec{E}}_{\text{Left}} = \underbrace{\rho}_{\text{Right}}$	Left side: Divergence of the electric field, $\vec{E}$ – the tendency of the field to “flow” away from a specified location. Right side: Electric charge density, ρ	The electric field, $\vec{E}$ produced by electric charge diverges from positive charge and converges upon negative charge. The electric field, $\vec{E}$ is tendency to propagate perpendicularly away from a surface charge.

Gauss's Law (Magnetic fields)	Integral form:	Description	Information
	$\underbrace{\mu_0 \oint \vec{H} \cdot d\vec{S}}_{\text{Left}} = \underbrace{0}_{\text{Right}}$	Left side: The number of magnetic field lines – perpendicularly passing through a closed surface. Right side: Identically zero.	The total magnetic flux passing through any closed surface is zero. Flux enter the closed surface is same with the flux come out from the surface.
	Differential form:	Description	Information
	$\underbrace{\mu_0 \vec{\nabla} \cdot \vec{H}}_{\text{Left}} = \underbrace{0}_{\text{Right}}$	Left side: Divergence of the magnetic field – the tendency of the field to “flow” away from a point than toward it. Right side: Identically zero.	The divergence of the magnetic field at any point is zero.



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Faraday's Law	Integral form: $\underbrace{\oint_C \vec{E} \cdot d\vec{l}}_{\text{Left}} = -\underbrace{\mu_o \int_S \frac{\partial \vec{H}}{\partial t} \cdot d\vec{S}}_{\text{Right}}$	Left side: The circulation of the vector electric field, $\vec{E}$ around a closed path, C . Right side: The rate of change with time (d/dt) of magnetic field, through any surface, $\vec{S}$.	Changing magnetic flux through a surface induces an emf in any boundary path, C of that surface, and a changing magnetic field, $\vec{H}$ induces a circulating electric field.
	Differential form: $\underbrace{\vec{\nabla} \times \vec{E}}_{\text{Left}} = -\underbrace{\mu_o \frac{\partial \vec{H}}{\partial t}}_{\text{Right}}$	Left side: Curl of the electric field, – the tendency of the field lines to circulate around a point. Right side: The rate of change of the magnetic field, $\vec{H}$ over time (d/dt)	A circulating electric field, is produced by a magnetic field, $\vec{H}$ that changes with time.
Ampere's Law	Integral form: $\underbrace{\oint_C \vec{H} \cdot d\vec{l}}_{\text{Left}} = \underbrace{\int_S \left(\vec{J}_c + \epsilon_o \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}}_{\text{Right}}$	Left side: The circulation of the magnetic field, $\vec{H}$ around a closed path, C . Right side: Two sources for the magnetic field, $\vec{H}$; a steady conduction current, $\vec{J}_c$ and a changing electric field, $\vec{E}$ through any surface, bounded by closed path, C .	An electric current or a changing electric flux through a surface produces a circulating magnetic field around any path, C that bounds that surface.
	Differential form: $\underbrace{\vec{\nabla} \times \vec{H}}_{\text{Left}} = \underbrace{\vec{J}_c + \epsilon_o \frac{\partial \vec{E}}{\partial t}}_{\text{Right}}$	Left side: Curl of the magnetic field, – the tendency of the field lines to circulate around a point. Right side: Two terms represent the electric current density, $\vec{J}_c$ and the time rate of change of the electric field, $\vec{E}$.	A circulating electric field, is produced by a magnetic field, $\vec{H}$ that changes with time. An electric current, or a changing electric field, through a surface produces a circulating magnetic field, $\vec{H}$ around any path that bounds that surface.



Poynting Vector and Power Flow in Electromagnetic Fields:

Electromagnetic waves can transport energy from one point to another point. The electric and magnetic field intensities associated with a travelling electromagnetic wave can be related to the rate of such energy transfer.

Let us consider Maxwell's Curl Equations:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Using vector identity

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot \nabla \times \vec{E} - \vec{E} \cdot \nabla \times \vec{H}$$

The above curl equations we can write

$$\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \mu H^2 \right), \quad \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \mu E^2 \right)$$

$$\vec{E} \cdot \vec{J} = \sigma E^2$$

And

In simple medium where ϵ, μ and σ are constant, we can write

$$\therefore \nabla \cdot (\vec{E} \times \vec{H}) = -\frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) - \sigma E^2$$

Applying Divergence theorem we can write,

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dV - \int_V \sigma E^2 dV$$

$$\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dV$$

represents the rate of change of energy stored in the electric and magnetic fields and the term $\int_V \sigma E^2 dV$

.....(a)

The term

represents the power dissipation within the volume. Hence right

hand side of the equation (a) represents the total decrease in power within the volume under consideration.



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$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} = \oint_S \vec{P} \cdot d\vec{S}$$

The left hand side of equation (6.36) can be written as $\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$ where

$$\vec{P} = \vec{E} \times \vec{H}$$

(W/m²) is called the Poynting vector and it represents the power density vector associated with the electromagnetic field. The integration of the Poynting vector over any closed surface gives the net power flowing out of the surface. Equation (6.36) is referred to as Poynting theorem and it states that the net power flowing out of a given volume is equal to the time rate of decrease in the energy stored within the volume minus the conduction losses.

Poynting vector for the time harmonic case:

For time harmonic case, the time variation is of the form $e^{j\omega t}$, and we have seen that instantaneous value of a quantity is the real part of the product of a phasor quantity and $e^{j\omega t}$ when $\cos \omega t$ is used as reference. For example, if we consider the phasor

$$\vec{E}(z) = \hat{a}_x E_x(z) = \hat{a}_x E_0 e^{-j\beta z}$$

then we can write the instantaneous field as

$$\vec{E}(z, t) = \text{Re} [\vec{E}(z) e^{j\omega t}] = E_0 \cos(\omega t - \beta z) \hat{a}_x$$

when E_0 is real.

Let us consider two instantaneous quantities A and B such that

$$A = \text{Re} [A e^{j\omega t}] = |A| \cos(\omega t + \alpha) \quad B = \text{Re} [B e^{j\omega t}] = |B| \cos(\omega t + \beta)$$

where A and B are the phasor quantities i.e.,

$$A = |A| e^{j\alpha}$$

$$B = |B| e^{j\beta}$$

Therefore,



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Since A and B are periodic
period , the time
value of the product form
denoted by $\overline{AB}$ can be
as

$$\begin{aligned} AB &= |A| \cos(\omega t + \alpha) |B| \cos(\omega t + \beta) \\ &= \frac{1}{2} |A| |B| [\cos(\alpha - \beta) + \cos(2\omega t + \alpha + \beta)] \\ T &= \frac{2\pi}{\omega} \end{aligned}$$

with
average
 AB ,
written

$$\begin{aligned} \overline{AB} &= \frac{1}{T} \int_0^T AB dt \\ \overline{AB} &= \frac{1}{2} |A| |B| \cos(\alpha - \beta) \end{aligned}$$

Further, considering the phasor quantities A and B , we find that

$$\begin{aligned} AB^* &= |A| e^{j\alpha} |B| e^{-j\beta} = |A| |B| e^{j(\alpha - \beta)} \\ \text{Re}(AB^*) &= |A| |B| \cos(\alpha - \beta) \end{aligned}$$

, where $*$ denotes complex conjugate.

$$\therefore \overline{AB} = \frac{1}{2} \text{Re}(AB^*)$$

and

The poynting vector $\vec{P} = \vec{E} \times \vec{H}$ can be expressed as

$$\begin{aligned} \vec{P} &= \hat{a}_x (E_y H_z - E_z H_y) + \hat{a}_y (E_z H_x - E_x H_z) + \hat{a}_z (E_x H_y - E_y H_x) \\ &\dots\dots\dots(b) \end{aligned}$$

If we consider a plane electromagnetic wave propagating in $+z$ direction and has only E_x component, from (b) we can write:

$$\vec{P}_z = E_x(z, t) H_y(z, t) \hat{a}_z$$

Using (6.41)



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where

$$\vec{P}_{zav} = \frac{1}{2} \text{Re} \left(E_x(z) H_y^*(z) \hat{a}_z \right) \quad , \text{ for the plane}$$

wave

under consideration.

For

$$\vec{P}_{zav} = \frac{1}{2} \text{Re} \left(E_x(z) \times H_y(z) \right) \quad \text{a general case, we}$$

can

$$\vec{E}(z) = E_x(z) \hat{a}_x \quad \text{and} \quad \vec{H}(z) = H_y(z) \hat{a}_y \quad \text{write}$$

$$\vec{P}_{av} = \frac{1}{2} \text{Re} \left(\vec{E} \times \vec{H}^* \right)$$

We can define a complex Poynting vector

and time average of the instantaneous Poynting vector is given by

$$\vec{S} = \frac{1}{2} \vec{E} \times \vec{H}^*$$

$$\vec{P}_{av} = \text{Re}(\vec{S})$$

Polarisation of plane wave:

The polarization of a plane wave can be defined as the orientation of the electric field vector as a function of time at a fixed point in space. For an electromagnetic wave, the specification of the orientation of the electric field is sufficient as the magnetic field components are related to electric field vector by the Maxwell's equations.

Let us consider a plane wave travelling in the +z direction. The wave has both E_x and E_y components.

$$\vec{E} = \left(\hat{a}_x E_{ox} + \hat{a}_y E_{oy} \right) e^{-j\beta z}$$

The corresponding magnetic fields are given by,

$$\begin{aligned} \vec{H} &= \frac{1}{\eta} \hat{a}_z \times \vec{E} \\ &= \frac{1}{\eta} \hat{a}_z \times \left(\hat{a}_x E_{ox} + \hat{a}_y E_{oy} \right) e^{-j\beta z} \\ &= \frac{1}{\eta} \left(-E_{oy} \hat{a}_x + E_{ox} \hat{a}_y \right) e^{-j\beta z} \end{aligned}$$

Depending upon the values of E_{ox} and E_{oy} we can have several possibilities:

1. If $E_{oy} = 0$, then the wave is linearly polarised in the x-direction.
2. If $E_{ox} = 0$, then the wave is linearly polarised in the y-direction.



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3. If E_{ox} and E_{oy} are both real (or complex with equal phase), once again we get a linearly polarised wave

with the axis of polarisation inclined at an angle $\tan^{-1} \frac{E_{oy}}{E_{ox}}$, with respect to the x-axis.

This is shown in fig 6.4.

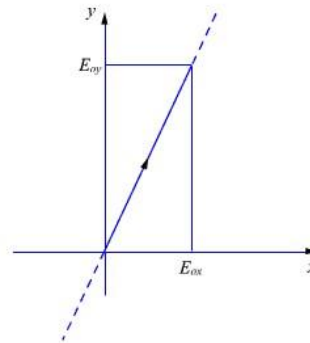


Fig 6.4 : Linear Polarisation

If E_{ox} and E_{oy} are complex with different phase angles, $\vec{E}$ will not point to a single spatial direction. This is explained as follows:

$$E_{ox} = |E_{ox}| e^{j\alpha} \quad E_{oy} = |E_{oy}| e^{j\beta}$$

$$E_x(z, t) = \text{Re} \left[|E_{ox}| e^{j\alpha} e^{-j\beta z} e^{j\omega t} \right] = |E_{ox}| \cos(\omega t - \beta z + \alpha)$$

$$E_y(z, t) = \text{Re} \left[|E_{oy}| e^{j\beta} e^{-j\beta z} e^{j\omega t} \right] = |E_{oy}| \cos(\omega t - \beta z + b)$$

Let

Then,

$$b = \frac{\pi}{2}$$

and

To keep the things simple, let us consider $\alpha = 0$ (c)

. Further, let us study the nature of the electric field on the $z = 0$ plane.

From equation (c) we find that,

$$E_x(0, t) = |E_{ox}| \cos \omega t$$

$$E_y(0, t) = |E_{oy}| \cos \left(\omega t + \frac{\pi}{2} \right) = |E_{oy}| (-\sin \omega t)$$

$$\therefore \left(\frac{E_x(0, t)}{|E_{ox}|} \right)^2 + \left(\frac{E_y(0, t)}{|E_{oy}|} \right)^2 = \cos^2 \omega t + \sin^2 \omega t = 1$$



and the electric field vector at $z = 0$ can be written as

$$\vec{E}(o, t) = |E_{ox}| \cos(\omega t) \hat{a}_x - |E_{oy}| \sin(\omega t) \hat{a}_y \quad \dots\dots\dots (d)$$

Assuming $|E_{ox}| > |E_{oy}|$ and the plot of $\vec{E}(o, t)$ for various values of t is shown in figure 6.5.

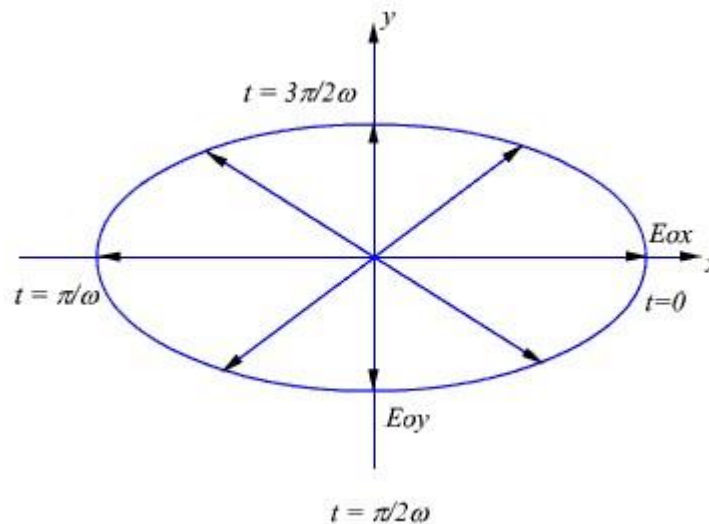


Figure 6.5 : Plot of $E(o, t)$

From equation (d) and figure (6.5) we observe that the tip of the arrow representing electric field vector traces an ellipse and the field is said to be elliptically polarized.

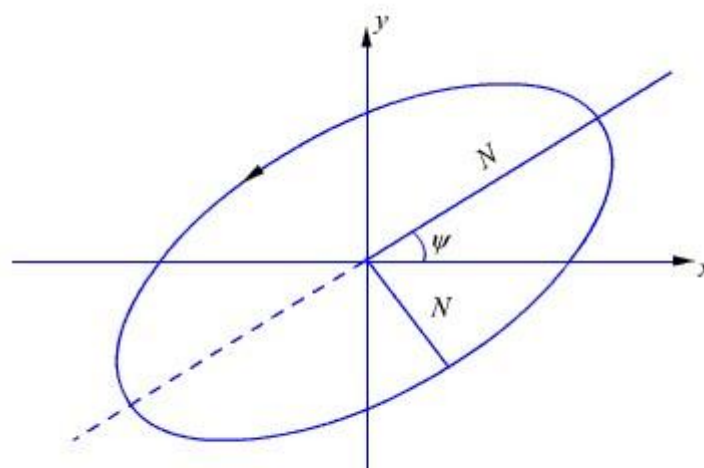


Figure 6.6: Polarisation ellipse



The polarisation ellipse shown in figure 6.6 is defined by its axial ratio(M/N , the ratio of semimajor to semiminor axis), tilt angle ψ (orientation with respect to xaxis) and sense of rotation(i.e., CW or CCW). Linear polarisation can be treated as a special case of elliptical polarisation, for which the axial ratio is infinite.

In our example, if $|E_{ox}| = |E_{oy}|$, from equation (6.47), the tip of the arrow representing electric field vector traces out a circle. Such a case is referred to as Circular Polarisation. For circular polarisation the axial ratio is unity.

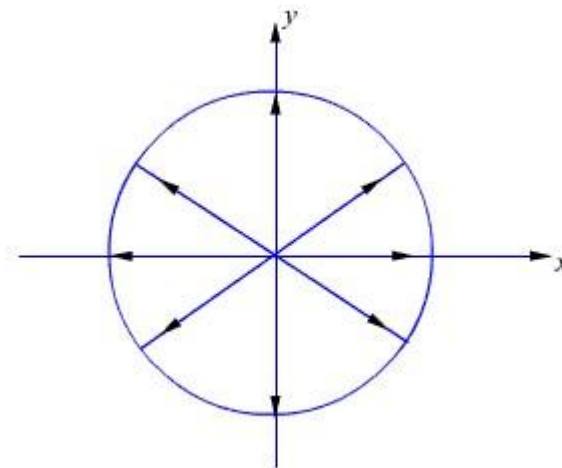


Figure 6.7: Circular Polarisation (RHCP)

Further, the circular polarisation is said to be right handed circular polarisation (RHCP) if the electric field vector rotates in the direction of the fingers of the right hand when the thumb points in the direction of propagation-(same and CCW). If the electric field vector rotates in the opposite direction, the polarisation is said to be left hand circular polarisation (LHCP) (same as CW).

In AM radio broadcast, the radiated electromagnetic wave is linearly polarised with the $\vec{E}$ field vertical to the ground(vertical polarisation) whereas TV signals are horizontally polarised waves. FM broadcast is usually carried out using circularly polarised waves.

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In radio communication, different information signals can be transmitted at the same frequency at orthogonal polarisation (one signal as vertically polarised other horizontally polarised or one as RHCP while the other as LHCP) to increase capacity. Otherwise, same signal can be transmitted at orthogonal polarisation to obtain diversity gain to improve reliability of transmission.